

From pest to pollinator

Using the mosquito transcriptome to investigate floral humidity as a foraging cue for insect pollinators

Massimiliano Jacob Spensley

Supervised by Dr. Heather Whitney and Prof. Davide Pisani



Overview

Mosquitoes as vectors for disease

Mosquito humidity sensing

Floral humidity and insect pollinators

Mosquitoes as pollinator models

Project plan

Overview

Mosquitoes as vectors for disease

Mosquito humidity sensing

Floral humidity and insect pollinators

Mosquitoes as pollinator models

Project plan

Mosquito-borne diseases

Mosquitoes are vectors for some of the deadliest infectious diseases worldwide...

Mosquito-borne diseases

Mosquitoes are vectors for some of the deadliest infectious diseases worldwide...



Aedes aegypti

Primary vector for **yellow fever**, **dengue**, **zika**, and **chikungunya** viruses.

Mosquito-borne diseases

Mosquitoes are vectors for some of the deadliest infectious diseases worldwide...

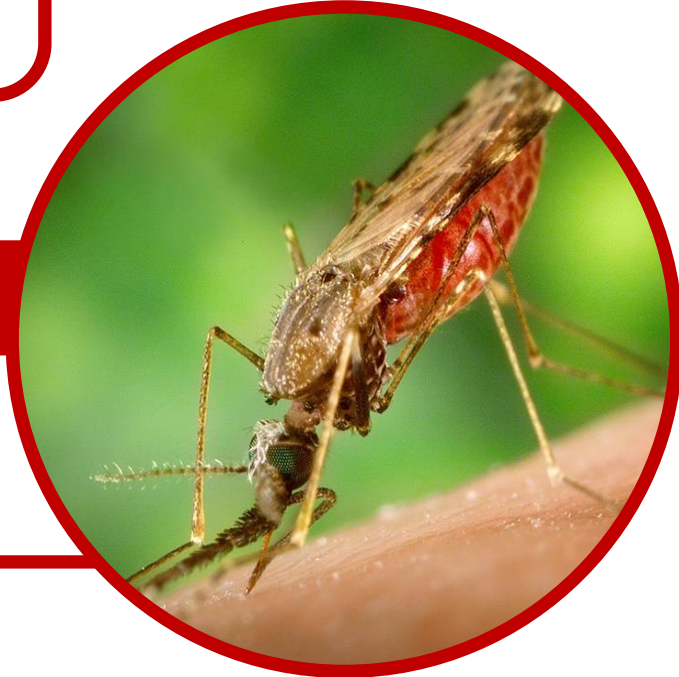


Aedes aegypti

Primary vector for **yellow fever**, **dengue**, **zika**, and **chikungunya** viruses.

Anopheles gambiae

Responsible for the majority of the global **malaria** burden.



Overview

Mosquitoes as vectors for disease

Mosquito humidity sensing

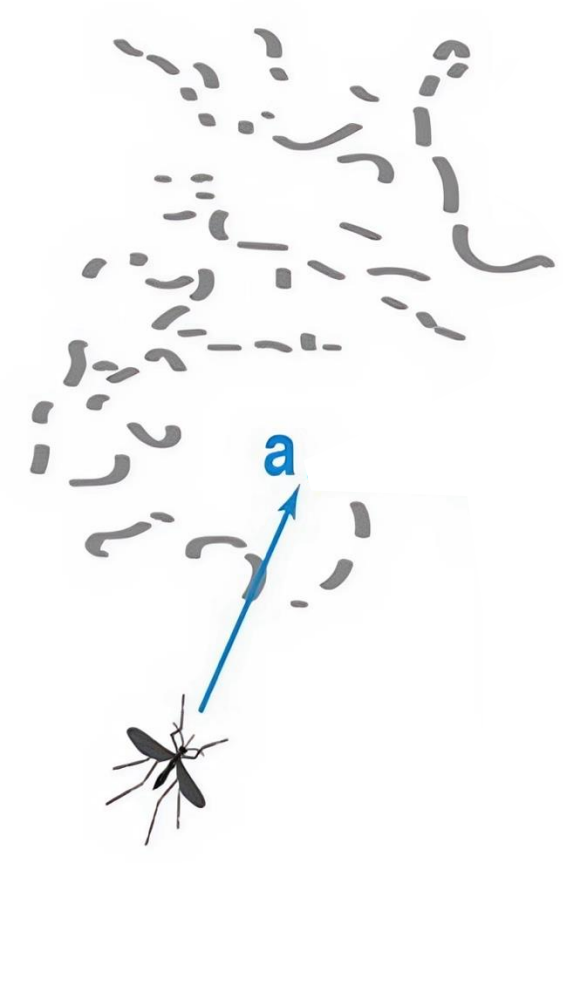
Floral humidity and insect pollinators

Mosquitoes as pollinator models

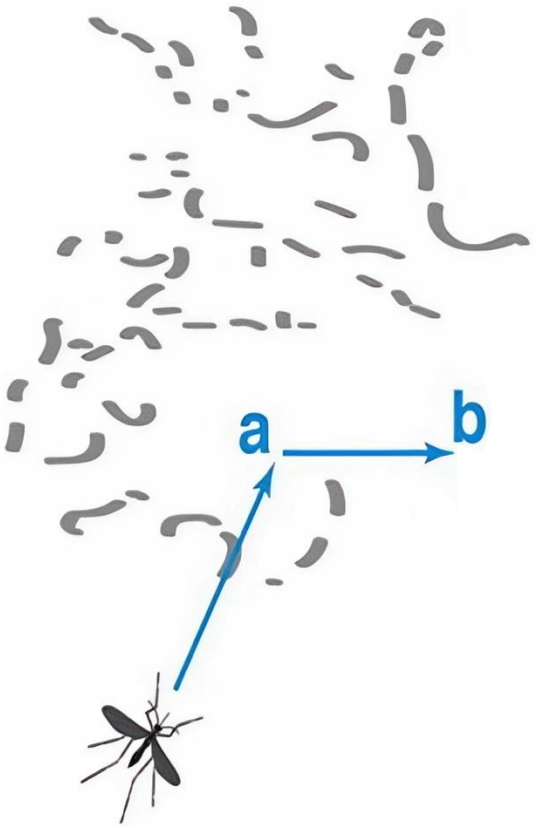
Project plan

Host location

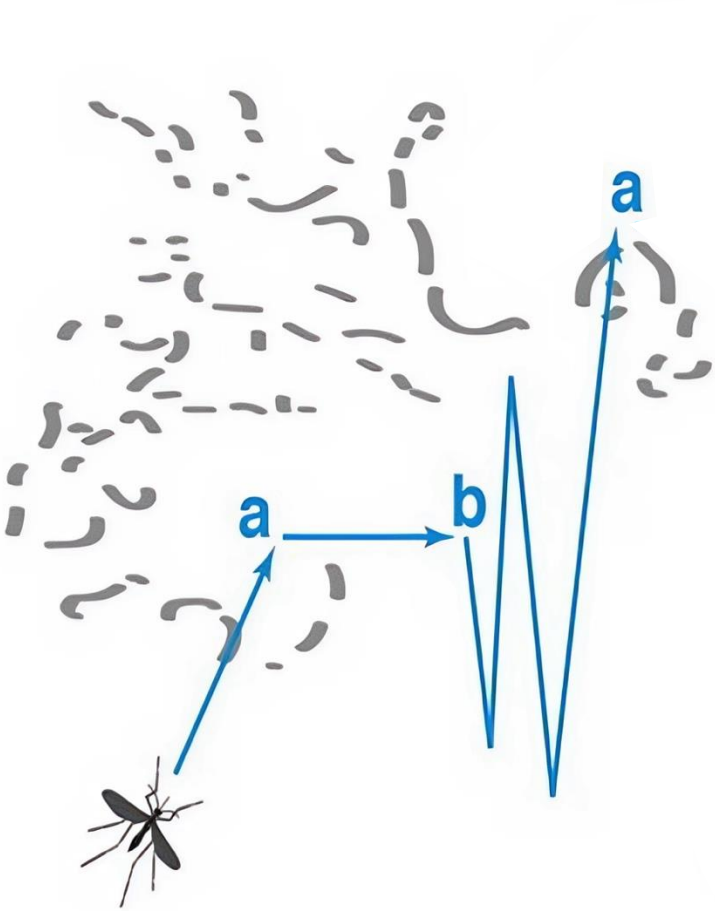
Host location



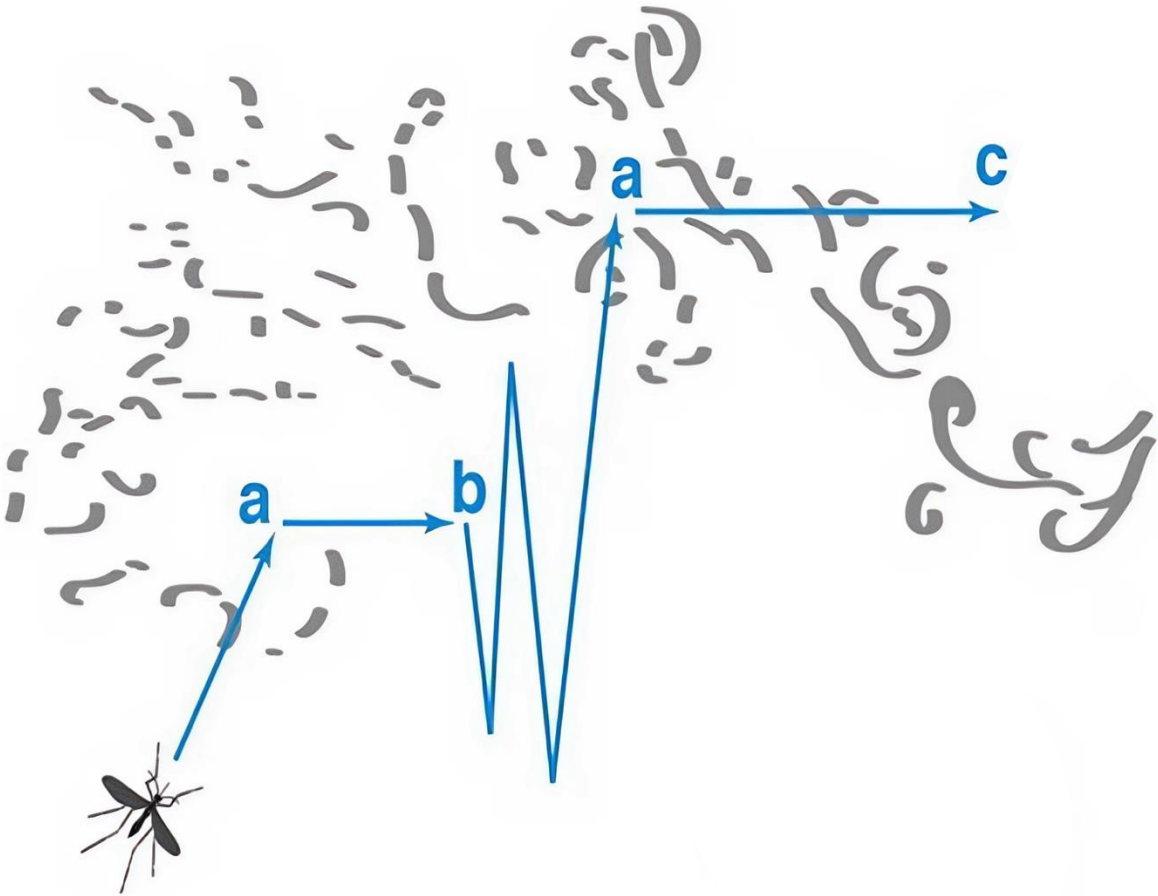
Host location



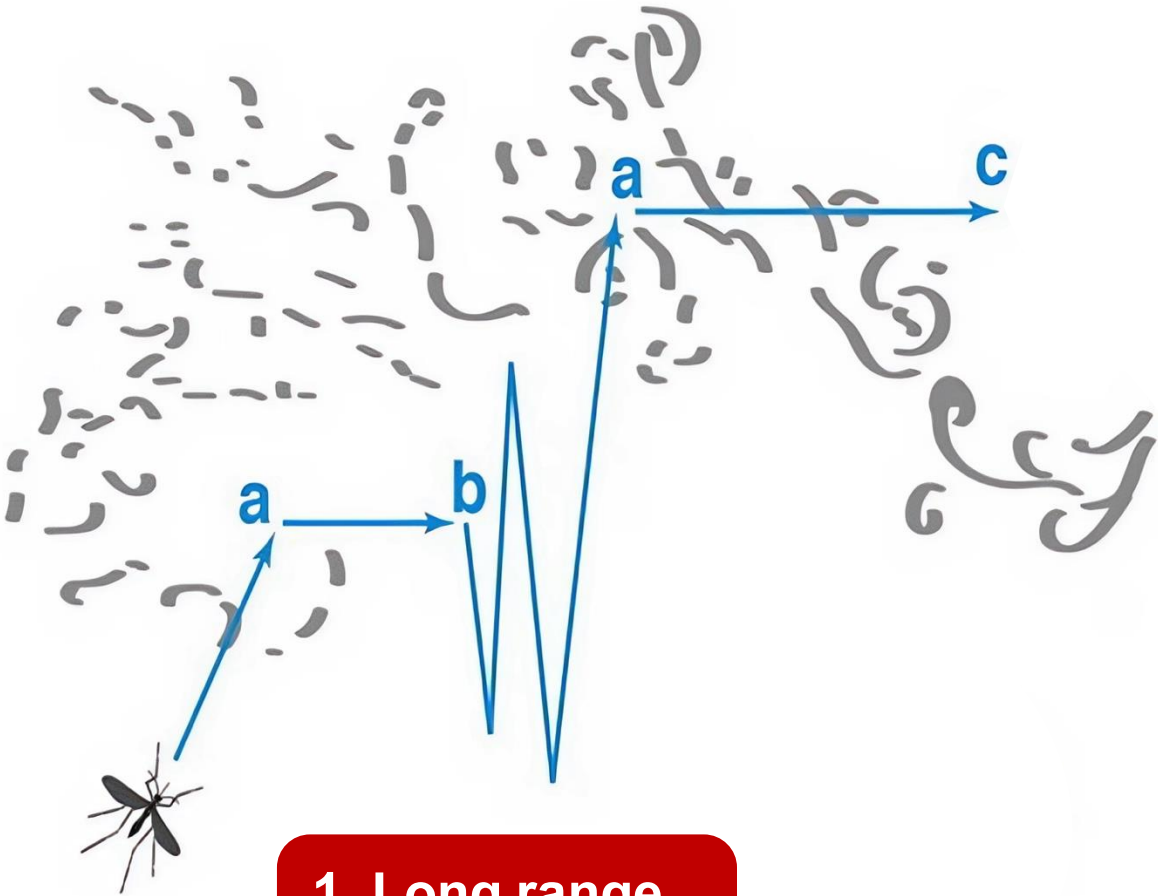
Host location



Host location



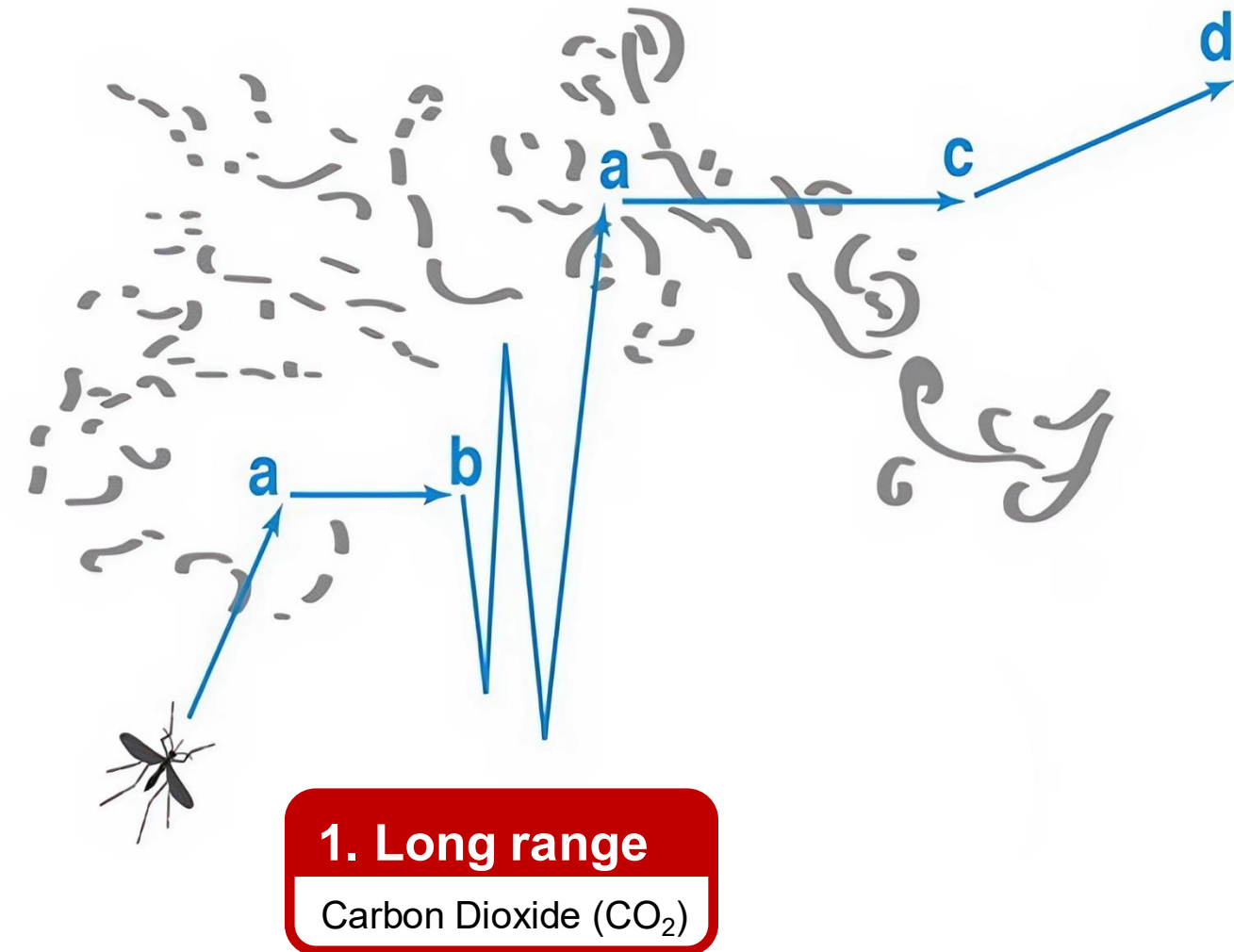
Host location



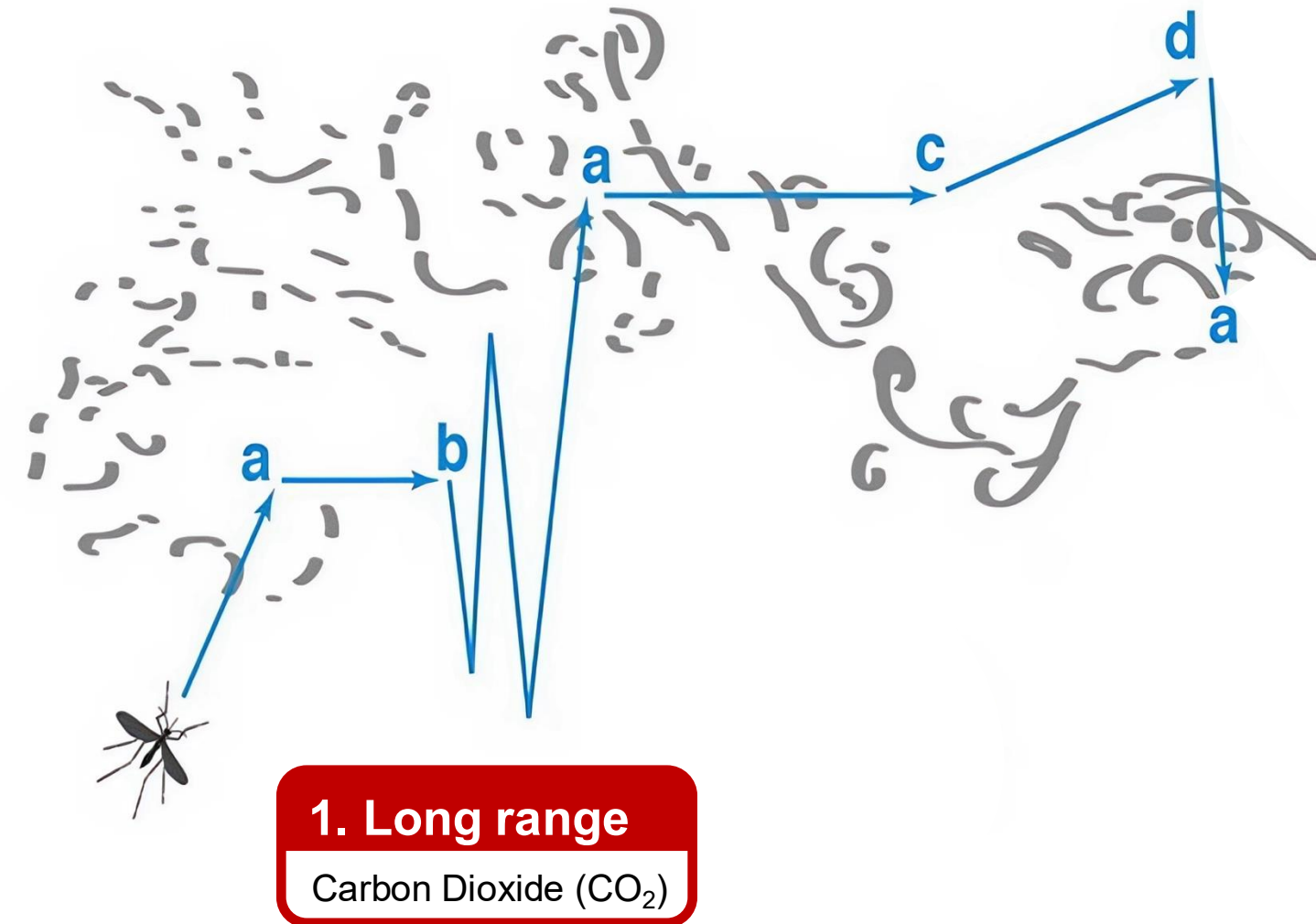
1. Long range

Carbon Dioxide (CO₂)

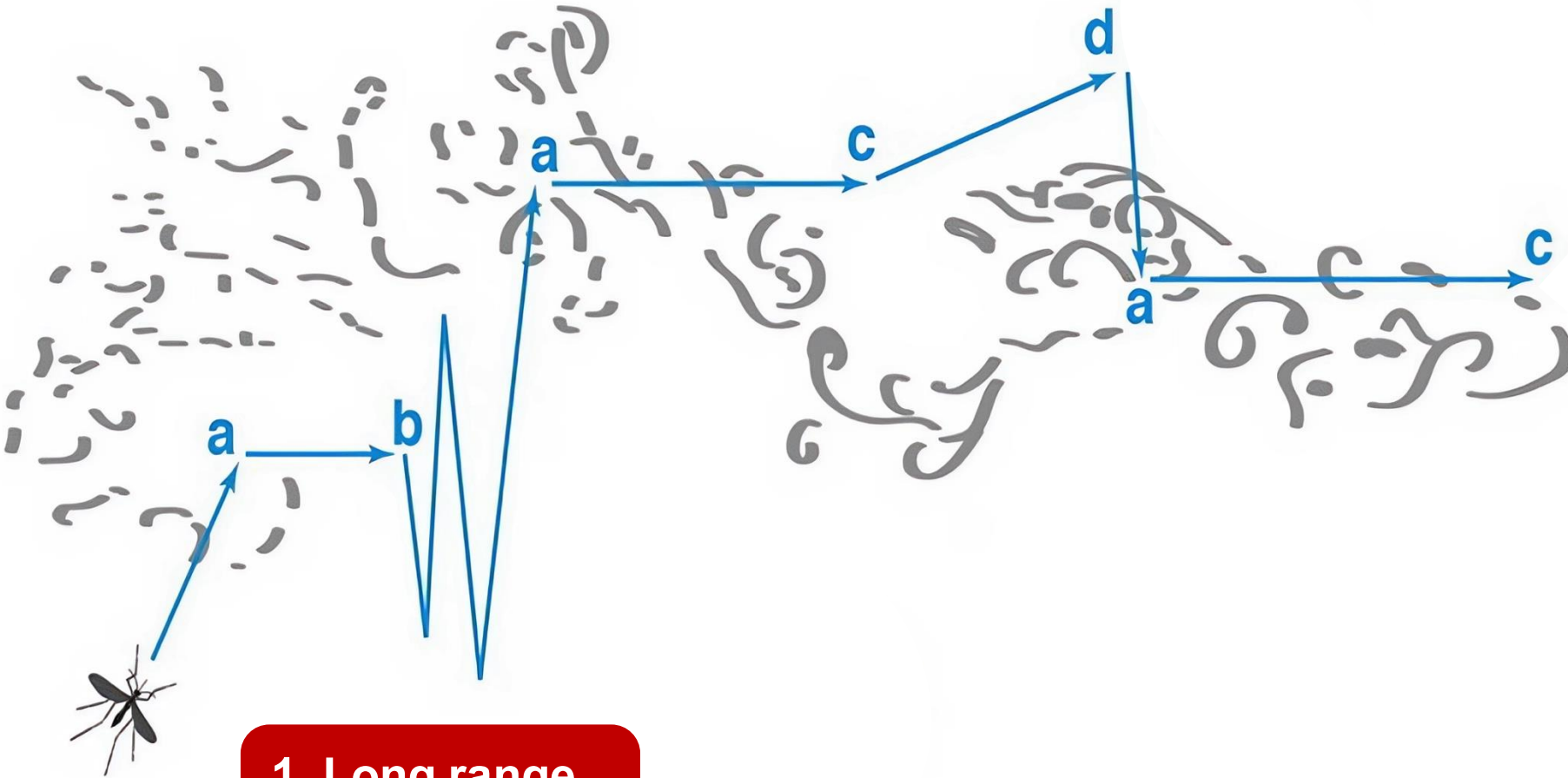
Host location



Host location



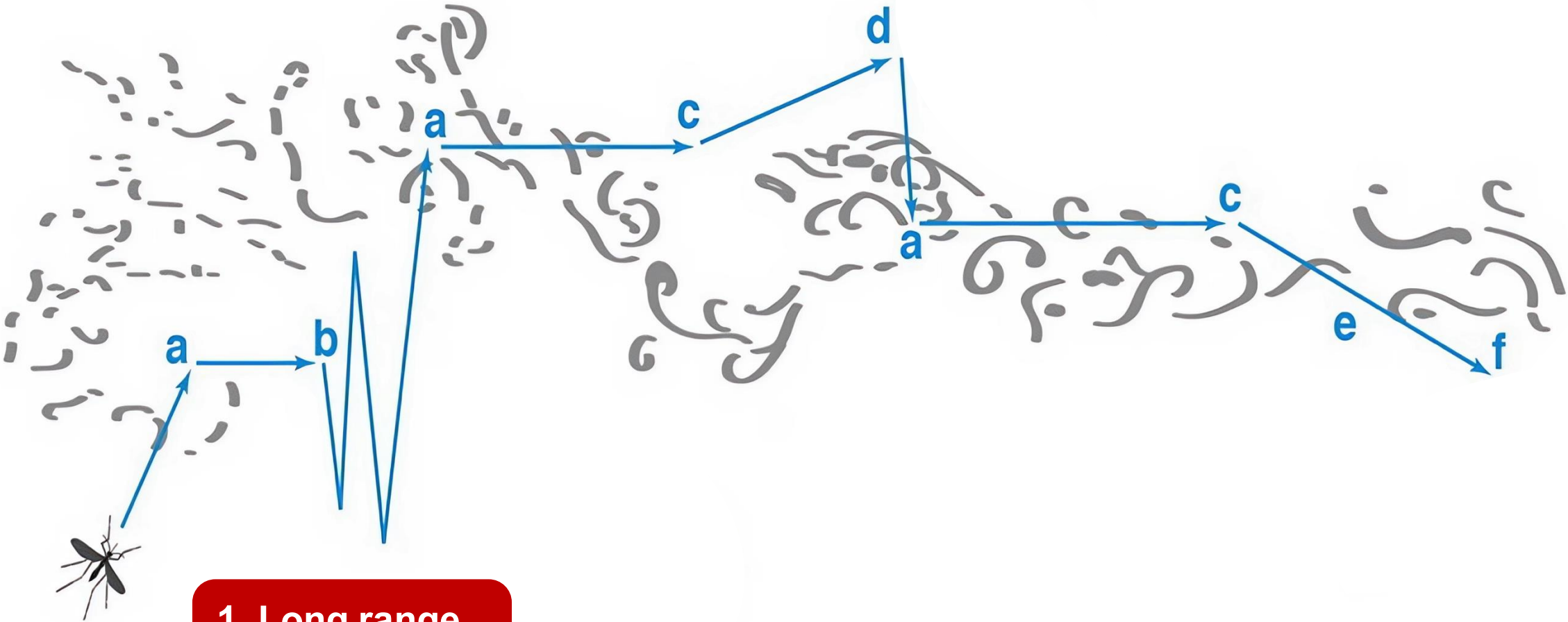
Host location



1. Long range

Carbon Dioxide (CO₂)

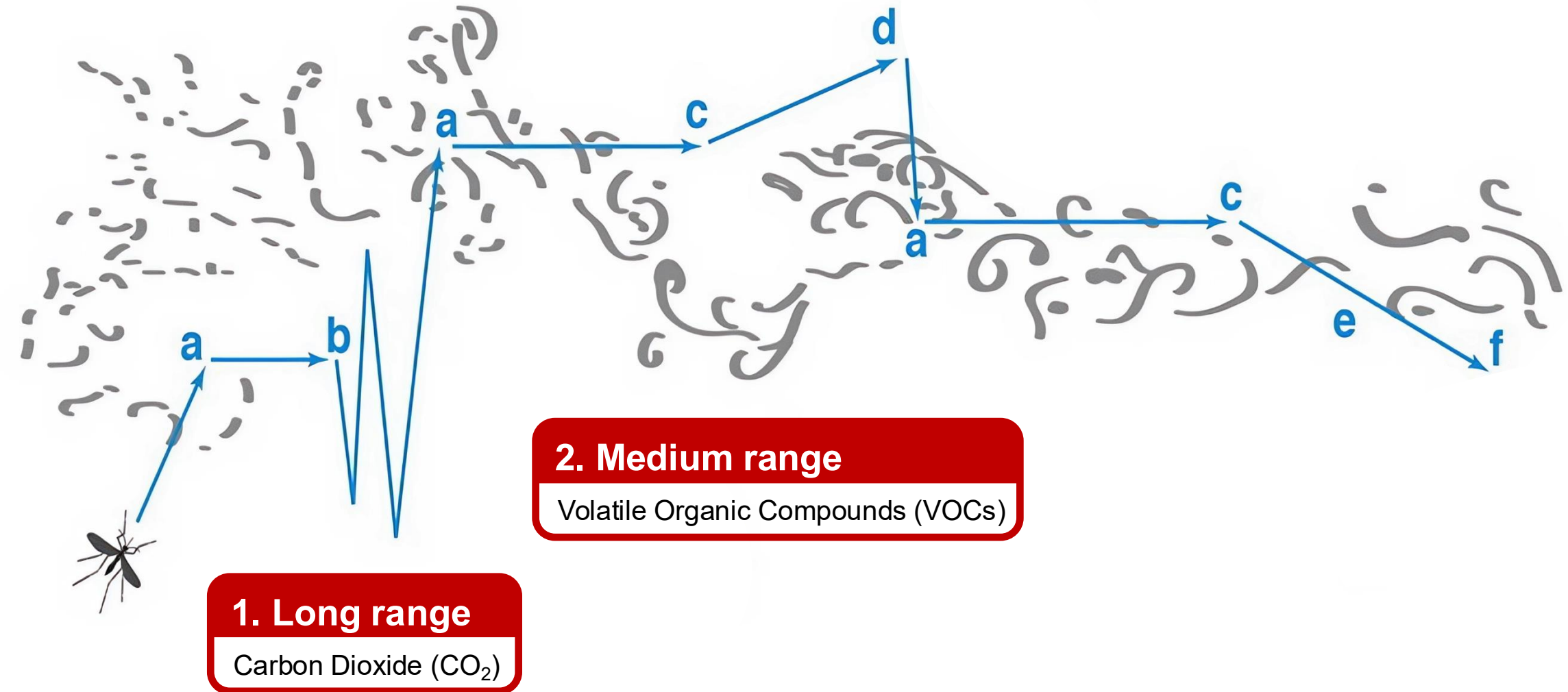
Host location



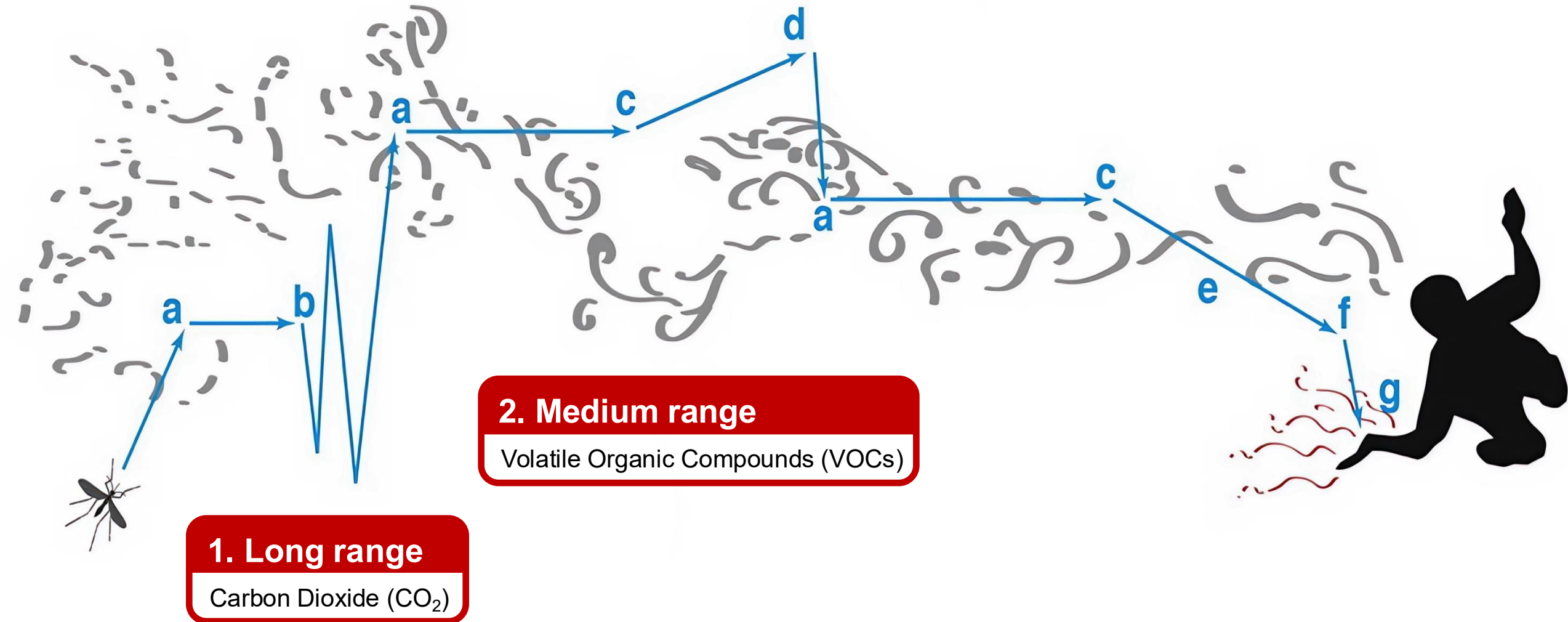
1. Long range

Carbon Dioxide (CO₂)

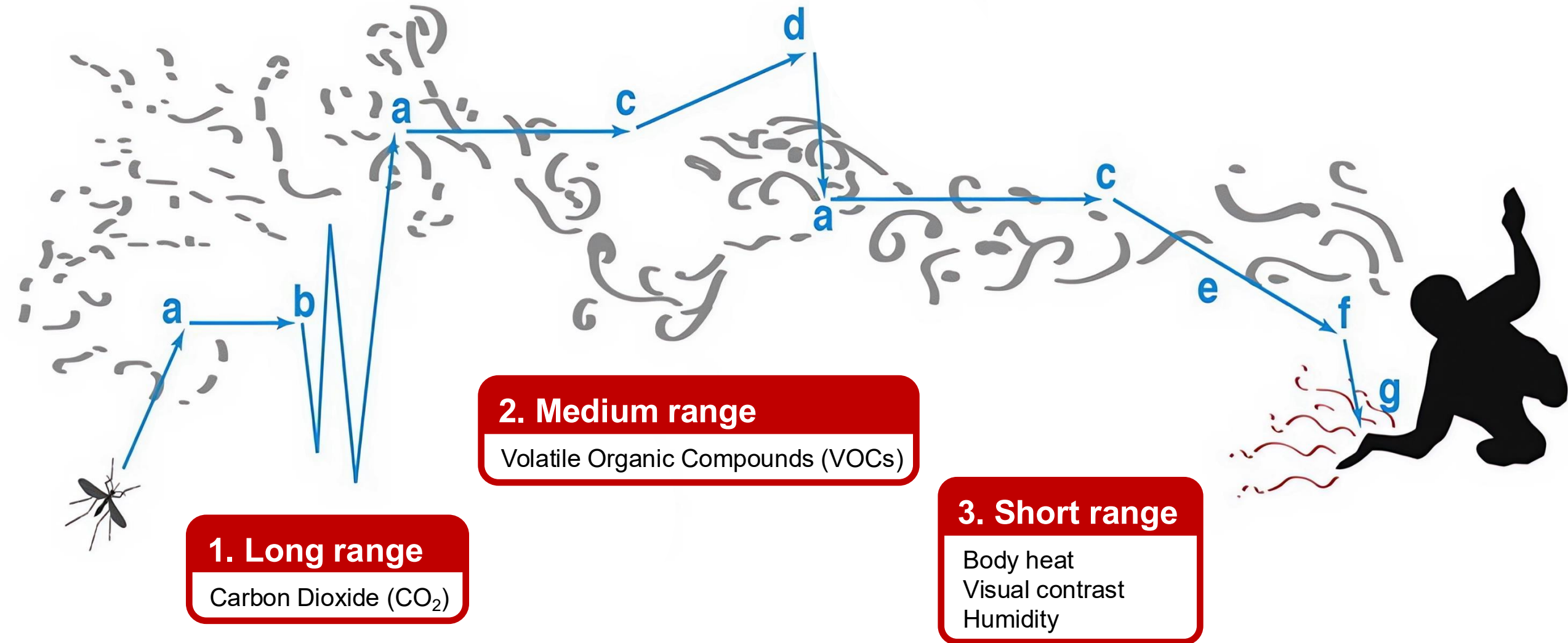
Host location



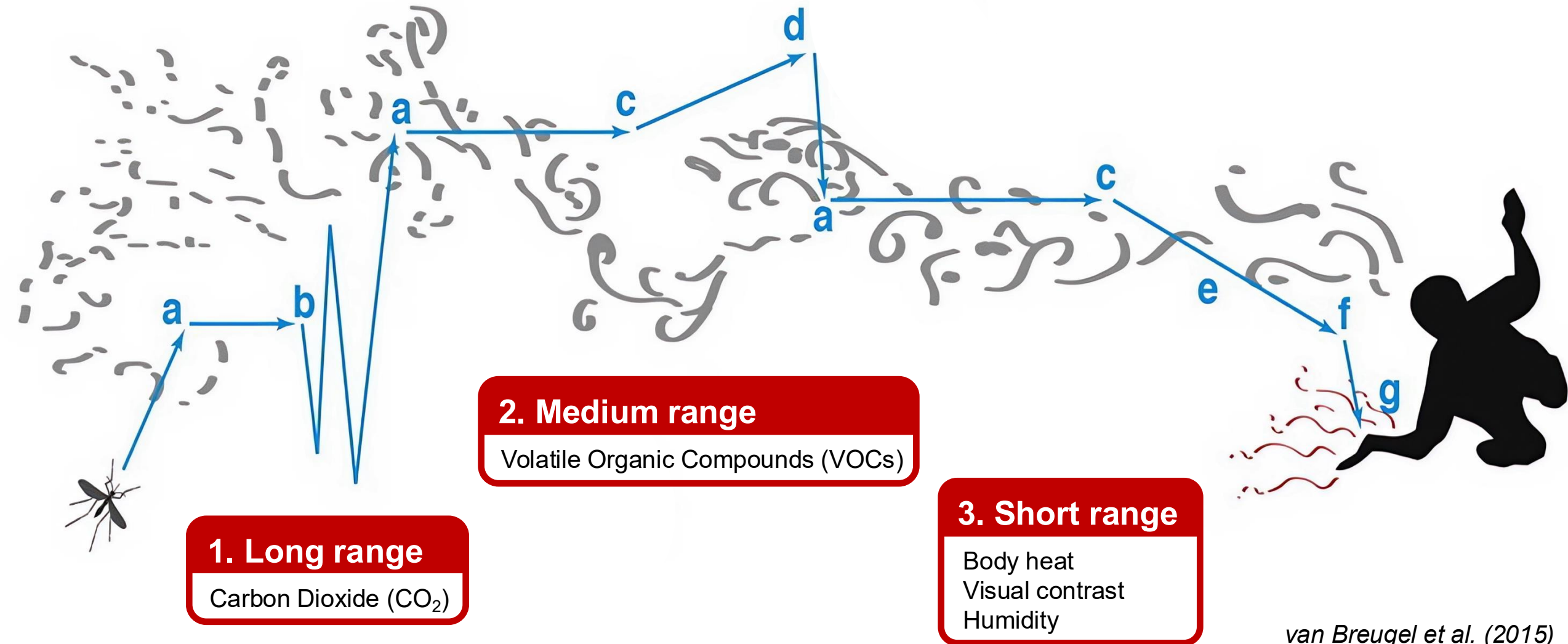
Host location



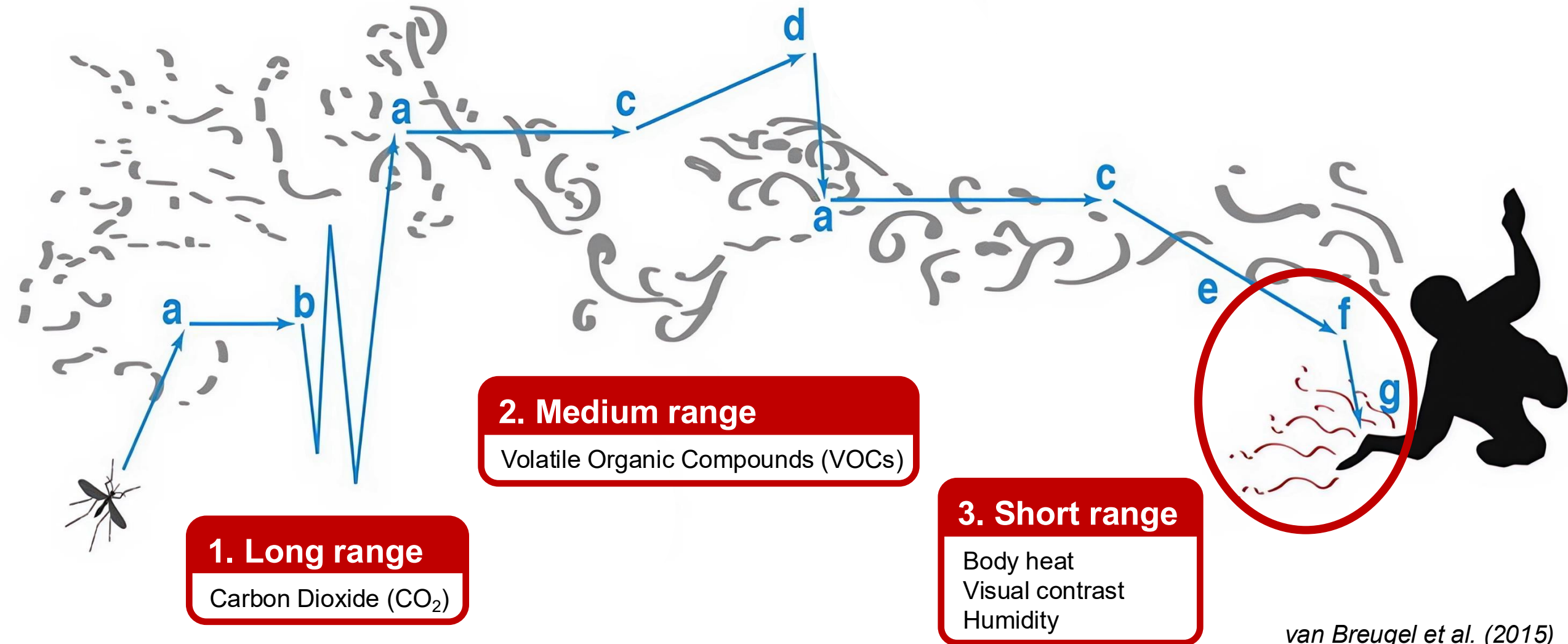
Host location



Host location



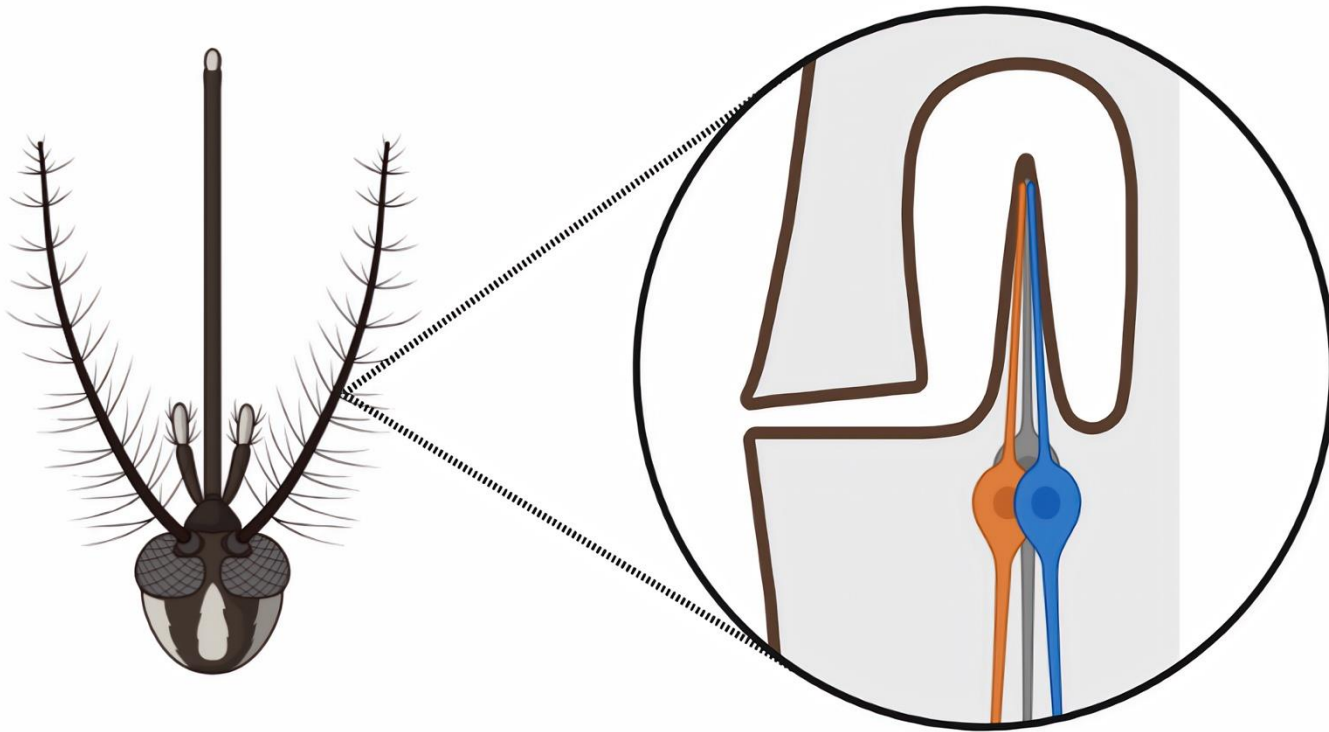
Host location



Humidity sensing in mosquitoes

Humidity sensing in mosquitoes

Hygrosensory sensilla ampullacea



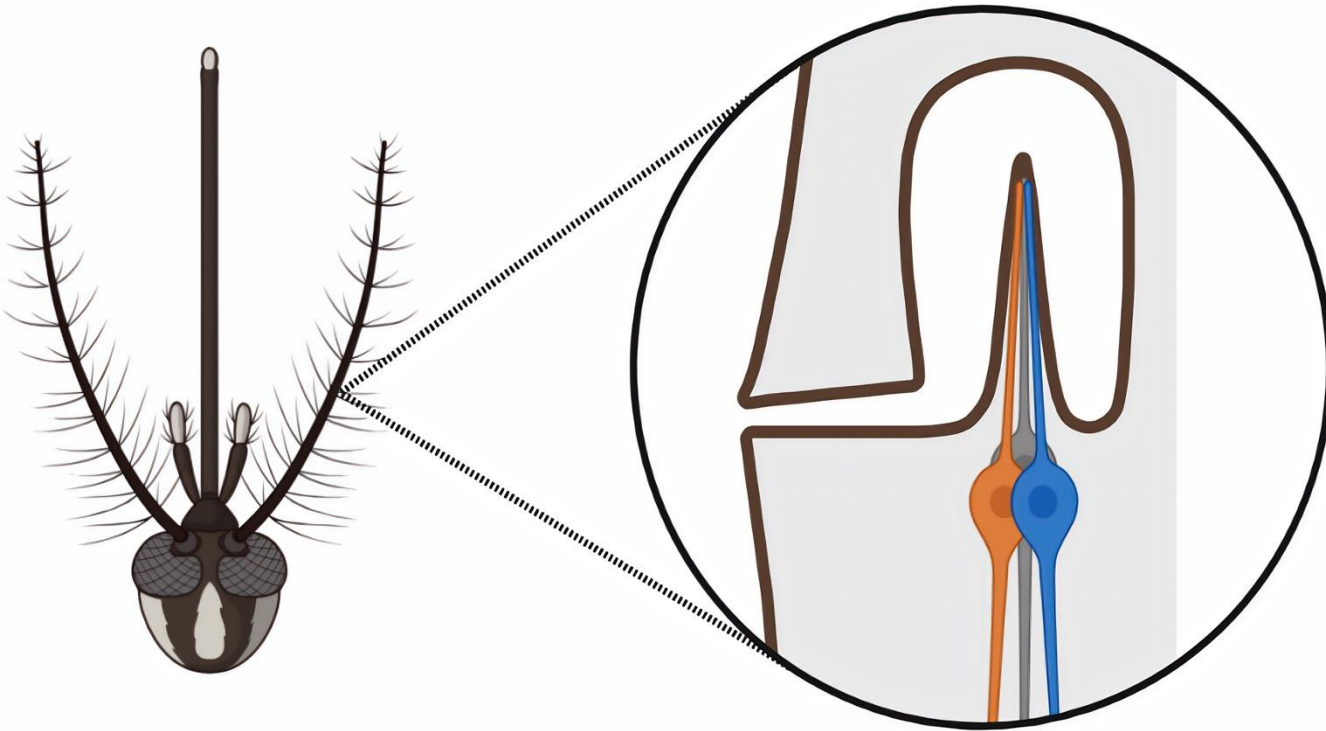
“Dry” cell



“Moist” cell

Humidity sensing in mosquitoes

Hygrosensory sensilla ampullacea



“Dry” cell



“Moist” cell

Ionotropic Receptors

Ir40a

Dry cells only.

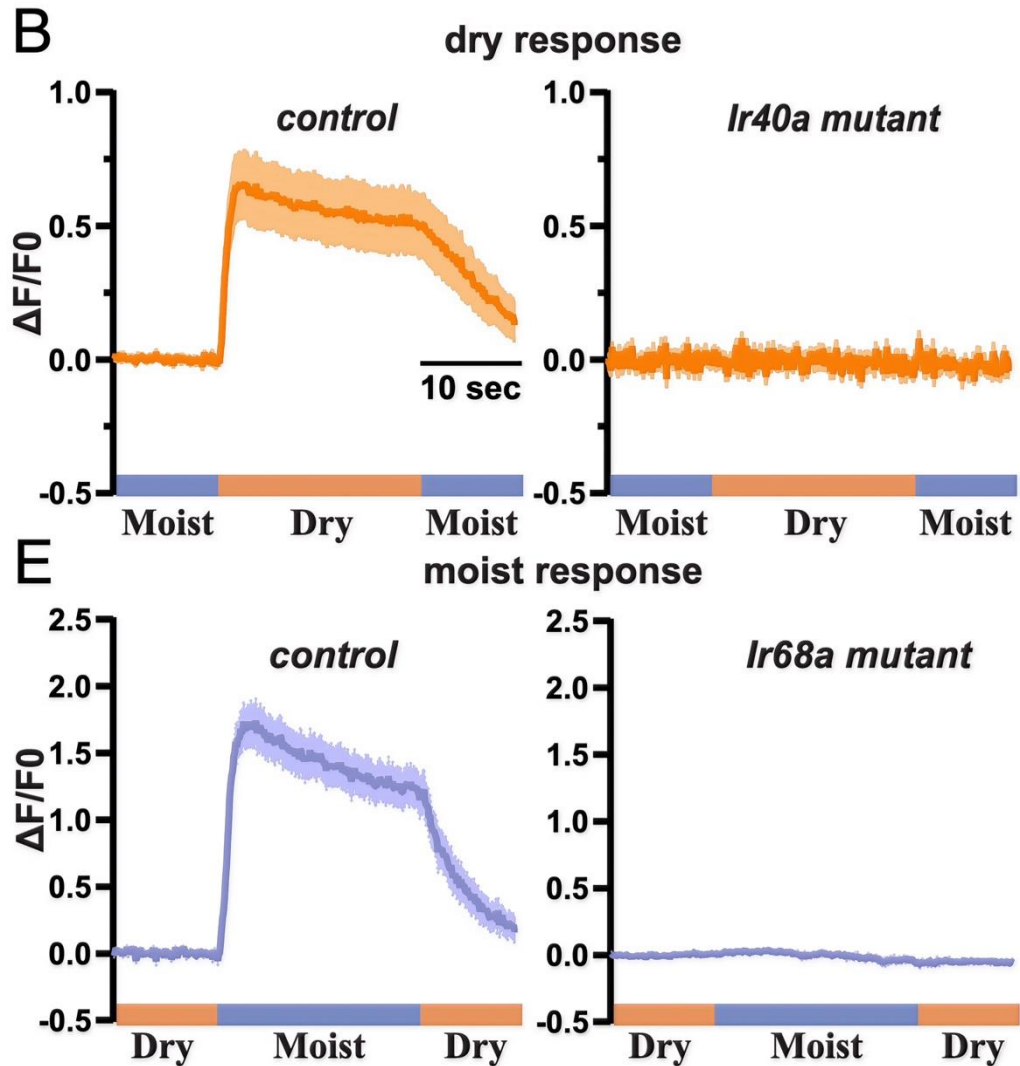
Ir68a

Moist cells only.

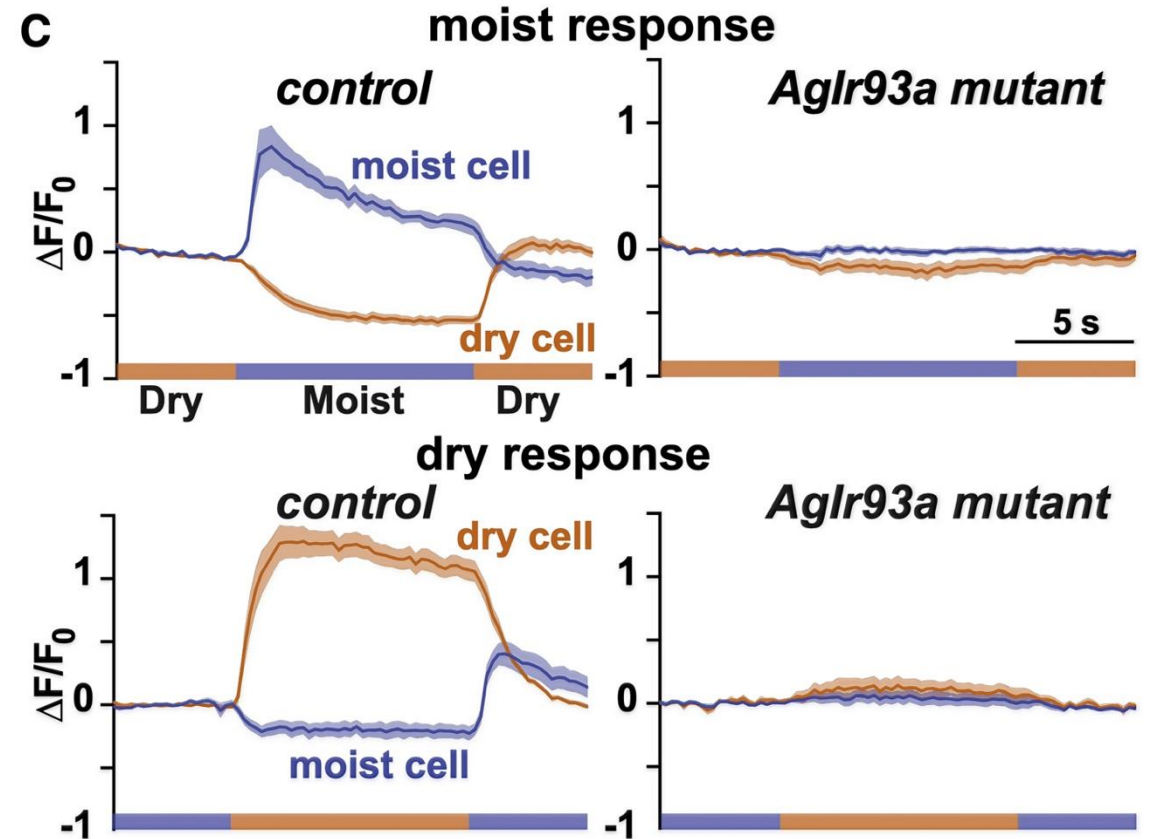
Ir93a

Both **dry** and **moist** cells.

Mosquito hygroreceptors



Tang et al. (2024)



Laursen et al. (2023)

Overview

Mosquitoes as vectors for disease

Mosquito humidity sensing

Floral humidity and insect pollinators

Mosquitoes as pollinator models

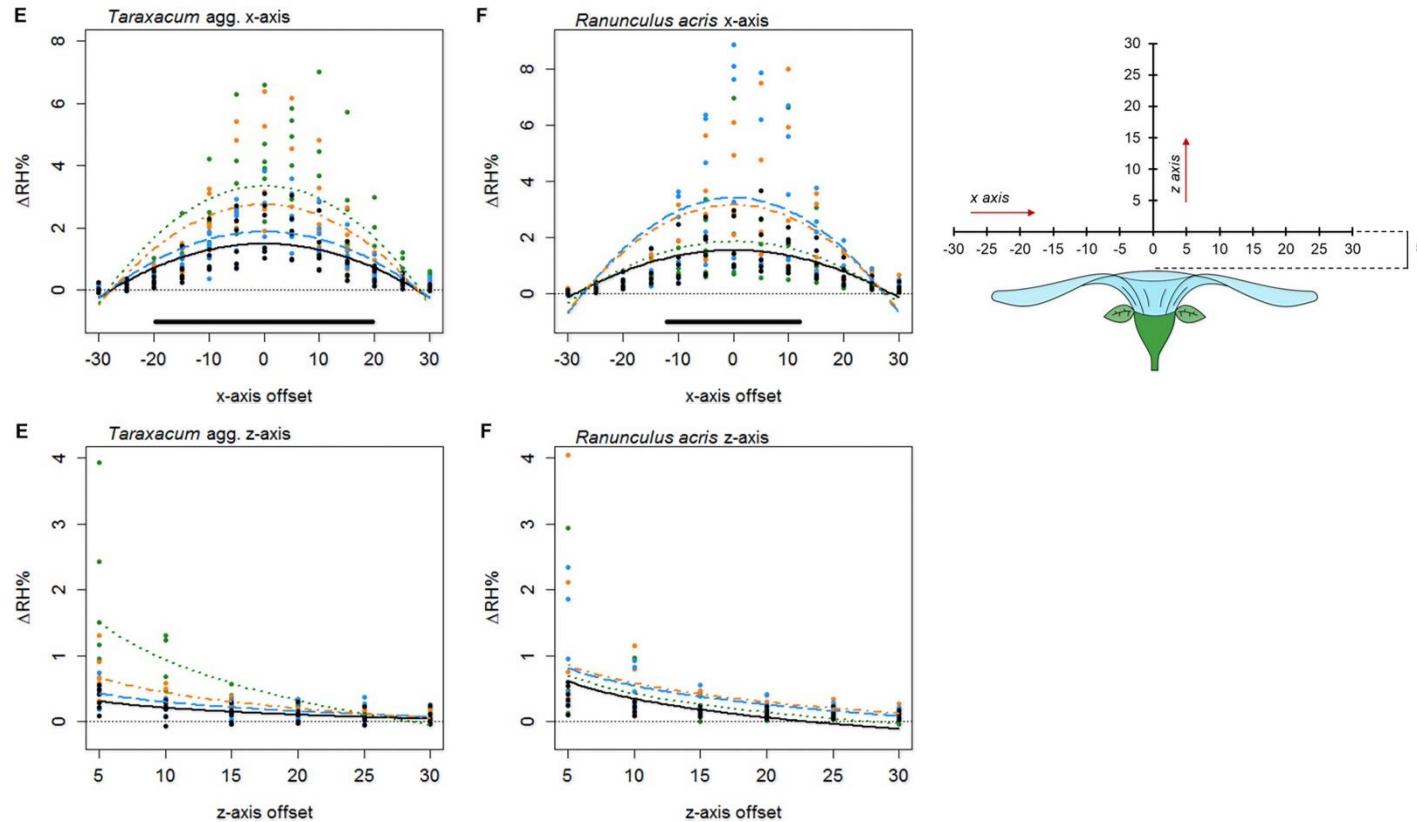
Project plan

Floral humidity

Humidity is elevated around the corolla of many flowering plants as a result of **nectar evaporation** and **petal/stomatal transpiration**.

Floral humidity

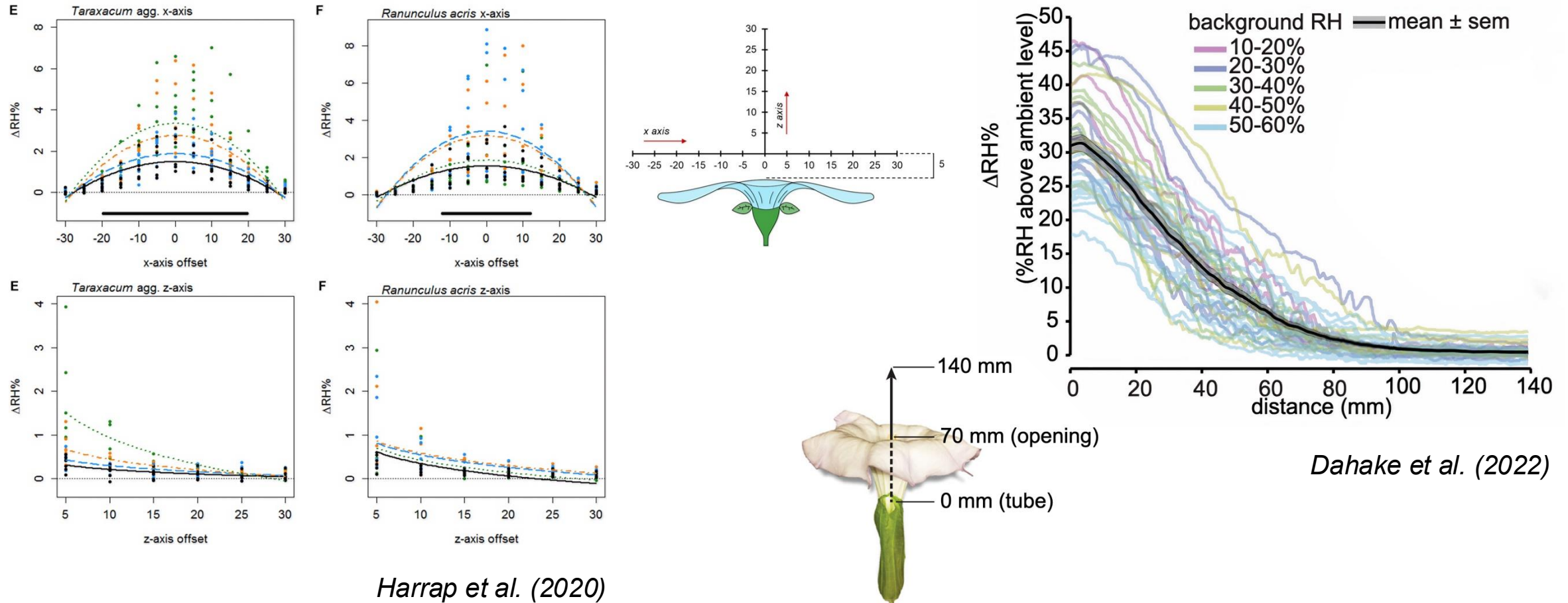
Humidity is elevated around the corolla of many flowering plants as a result of **nectar evaporation** and **petal/stomatal transpiration**.



Harrap et al. (2020)

Floral humidity

Humidity is elevated around the corolla of many flowering plants as a result of **nectar evaporation** and **petal/stomatal transpiration**.



Floral humidity as a foraging cue

Floral humidity as a reliable sensory cue for profitability assessment by nectar-foraging hawkmoths

Martin von Arx^{a,b,1}, Joaquín Goyret^a, Goggy Davidowitz^b, and Robert A. Raguso^a

^aDepartment of Neurobiology and Behavior, Cornell University, Ithaca, NY 14853; and ^bDepartment of Entomology, University of Arizona, Tucson, AZ 85721

Edited by John G. Hildebrand, University of Arizona, Tucson, AZ, and approved May 1, 2012 (received for review January 12, 2012)

Most research on plant–pollinator communication has focused on sensory and behavioral responses to relatively static cues. Floral rewards such as nectar, however, are dynamic, and foraging animals will increase their energetic profit if they can make use of floral cues that more accurately indicate nectar availability. Here we document such a cue—transient humidity gradients—using the night blooming flowers of *Oenothera cespitosa* (Onagraceae). The headspace of newly opened flowers reaches levels of about 4% above ambient relative humidity due to additive evapotranspirational water loss through petals and water-saturated air from the nectar tube. Floral humidity plumes differ from ambient levels only during the first 30 min after anthesis (before nectar is depleted in wild populations), whereas other floral traits (scent, shape, and color) persist for 12–24 h. Manipulative experiments indicated that floral humidity gradients are mechanistically linked to nectar volume and therefore contain information about energy rewards to floral visitors. Behavioral assays with *Hyles lineata* (Sphingidae) and artificial flowers with appropriate humidity gradients suggest that these hawkmoth pollinators distinguish between subtle differences in relative humidity when other floral cues are held constant. Moths consistently approached and probed flowers with elevated humidity over those with ambient humidity levels. Because floral humidity gradients are largely produced by the evaporation of nectar itself, they represent condition-informative cues that facilitate remote sensing of floral profitability by discriminating foragers. In a xeric environment, this level of honest communication should be adaptive when plant reproductive success is pollinator limited, due to intense competition for the attention of a specialized pollinator.

pollination biology | honest signaling | floral display

Plant–pollinator relationships—like all mutualisms—strike a tenuous balance between the conflicting interests of foraging animals and flowering plants (1, 2). For flowering plants, pollinator attraction through floral density, form, color, pattern, and odor is necessary but insufficient to ensure effective pollination. Ideally, pollinators should transfer pollen between conspecific plants in ways that result in favorable levels of outcrossing (3) and reduced pollen loss (4). Flowers commonly manipulate pollinator movement by varying quality or quantity of floral nectar. The offer of variable amounts of nectar (5, 6), toxic nectar (7, 8), or an empty promise of nectar elicit similar results: pollinators leave plants sooner, reducing for example geitonogamous inbreeding (if self-compatible) or pollen discounting (if self-incompatible).

The availability of floral rewards often coincides with floral display and pollinator activity on a gross temporal scale (i.e., hours) (9, 10). Nevertheless, most pollinators encounter reward-

© 2021. Published by The Company of Biologists Ltd | Journal of Experimental Biology (2021) 224, jeb240861. doi:10.1242/jeb.240861

RESEARCH ARTICLE

Bumblebees can detect floral humidity

Michael J. M. Harrap^{1,2,*}, Natalie Hempel de Ibarra², Henry D. Knowles^{1,3}, Heather M. Whitney¹ and Sean A. Rands¹

ABSTRACT

Floral humidity, a region of elevated humidity in the headspace of the flower, occurs in many plant species and may add to their multimodal floral displays. So far, the ability to detect and respond to floral humidity cues has been only established for hawkmoths when they locate and extract nectar while hovering in front of some moth-pollinated flowers. To test whether floral humidity can be used by other more widespread generalist pollinators, we designed artificial flowers that presented biologically relevant levels of humidity similar to those shown by flowering plants. Bumblebees showed a spontaneous preference for flowers that produced higher floral humidity. Furthermore, learning experiments showed that bumblebees are able to use differences in floral humidity to distinguish between rewarding and non-rewarding flowers. Our results indicate that bumblebees are sensitive to different levels of floral humidity. In this way floral humidity can add to the information provided by flowers and could impact pollinator behaviour more significantly than previously thought.

KEY WORDS: Behaviour, Floral Display, Multimodal cues, Learning, Bumblebees, Angiosperms, Flower, Pollination

INTRODUCTION

Floral humidity, an area of elevated humidity within the headspace of the flower, has been demonstrated to occur in a number of flower species (Corbet et al., 1979; Nordström et al., 2017; von Arx et al., 2012). Floral humidity is created by a combination of nectar evaporation and floral transpiration (Azad et al., 2007; Corbet et al., 1979; Harrap et al., 2020a; von Arx et al., 2012), although the contribution of these two influences may vary between species. Transects of the flower headspace of 42 species found 30 (71%) produce floral humidity of an intensity greater than would be

nature communications

Article

<https://doi.org/10.1038/s41467-022-35353-8>

A signal-like role for floral humidity in a nocturnal pollination system

Received: 17 April 2022

Accepted: 29 November 2022

Published online: 15 December 2022

 Check for updates

Ajinkya Dahake^{1✉}, Piyush Jain², Caleb C. Vogt¹, William Kandalaft¹, Abraham D. Stroock³ & Robert A. Raguso¹

Previous studies have considered floral humidity to be an inadvertent consequence of nectar evaporation, which could be exploited as a cue by nectar-seeking pollinators. By contrast, our interdisciplinary study of a night-blooming flower, *Datura wrightii*, and its hawkmoth pollinator, *Manduca sexta*, reveals that floral relative humidity acts as a mutually beneficial signal in this system. The distinction between cue- and signal-based functions is illustrated by three experimental findings. First, floral humidity gradients in *Datura* are nearly ten-fold greater than those reported for other species, and result from

von Arx et al. (2012)

Harrap et al. (2021)

Dahake et al. (2022)

Floral humidity as a foraging cue

Floral humidity as a reliable sensory cue for profitability assessment by nectar-foraging hawkmoths

Martin von Arx^{a,b,1}, Joaquín Goyret^a, Goggy Davidowitz^b, and Robert A. Raguso^a

^aDepartment of Neurobiology and Behavior, Cornell University, Ithaca, NY 14853; and ^bDepartment of Entomology, University of Arizona, Tucson, AZ 85721

Edited by John G. Hildebrand, University of Arizona, Tucson, AZ, and approved May 1, 2012 (received for review January 12, 2012)

Most research on plant–pollinator communication has focused on sensory and behavioral responses to relatively static cues. Floral rewards such as nectar, however, are dynamic, and foraging animals will increase their energetic profit if they can make use of floral cues that more accurately indicate nectar availability. Here we document such a cue—transient humidity gradients—using the night blooming flowers of *Oenothera cespitosa* (Onagraceae). The headspace of newly opened flowers reaches levels of about 4% above ambient relative humidity due to additive evapotranspirational water loss through petals and water-saturated air from the nectar tube. Floral humidity plumes differ from ambient levels only during the first 30 min after anthesis (before nectar is depleted in wild populations), whereas other floral traits (scent, shape, and color) persist for 12–24 h. Manipulative experiments indicated that floral humidity gradients are mechanistically linked to nectar volume and therefore contain information about energy rewards to floral visitors. Behavioral assays with *Hyles lineata* (Sphingidae) and artificial flowers with appropriate humidity gradients suggest that these hawkmoth pollinators distinguish between subtle differences in relative humidity when other floral cues are held constant. Moths consistently approached and probed flowers with elevated humidity over those with ambient humidity levels. Because floral humidity gradients are largely produced by the evaporation of nectar itself, they represent condition-informative cues that facilitate remote sensing of floral profitability by discriminating foragers. In a xeric environment, this level of honest communication should be adaptive when plant reproductive success is pollinator limited, due to intense competition for the attention of a specialized pollinator.

pollination biology | honest signaling | floral display

Plant–pollinator relationships—like all mutualisms—strike a tenuous balance between the conflicting interests of foraging animals and flowering plants (1, 2). For flowering plants, pollinator attraction through floral density, form, color, pattern, and odor is necessary but insufficient to ensure effective pollination. Ideally, pollinators should transfer pollen between conspecific plants in ways that result in favorable levels of outcrossing (3) and reduced pollen loss (4). Flowers commonly manipulate pollinator movement by varying quality or quantity of floral nectar. The offer of variable amounts of nectar (5, 6), toxic nectar (7, 8), or an empty promise of nectar elicit similar results: pollinators leave plants sooner, reducing for example geitonogamous inbreeding (if self-compatible) or pollen discounting (if self-incompatible).

The availability of floral rewards often coincides with floral display and pollinator activity on a gross temporal scale (i.e., hours) (9, 10). Nevertheless, most pollinators encounter reward-

© 2021. Published by The Company of Biologists Ltd | Journal of Experimental Biology (2021) 224, jeb240861. doi:10.1242/jeb.240861

RESEARCH ARTICLE

Bumblebees can detect floral humidity

Michael J. M. Harrap^{1,2,*}, Natalie Hempel de Ibarra², Henry D. Knowles^{1,3}, Heather M. Whitney¹ and Sean A. Rands¹

ABSTRACT

Floral humidity, a region of elevated humidity in the headspace of the flower, occurs in many plant species and may add to their multimodal floral displays. So far, the ability to detect and respond to floral humidity cues has been only established for hawkmoths when they locate and extract nectar while hovering in front of some moth-pollinated flowers. To test whether floral humidity can be used by other more widespread generalist pollinators, we designed artificial flowers that presented biologically relevant levels of humidity similar to those shown by flowering plants. Bumblebees showed a spontaneous preference for flowers that produced higher floral humidity. Furthermore, learning experiments showed that bumblebees are able to use differences in floral humidity to distinguish between rewarding and non-rewarding flowers. Our results indicate that bumblebees are sensitive to different levels of floral humidity. In this way floral humidity can add to the information provided by flowers and could impact pollinator behaviour more significantly than previously thought.

KEY WORDS: Behaviour, Floral Display, Multimodal cues, Learning, Bumblebees, Angiosperms, Flower, Pollination

INTRODUCTION

Floral humidity, an area of elevated humidity within the headspace of the flower, has been demonstrated to occur in a number of flower species (Corbet et al., 1979; Nordström et al., 2017; von Arx et al., 2012). Floral humidity is created by a combination of nectar evaporation and floral transpiration (Azad et al., 2007; Corbet et al., 1979; Harrap et al., 2020a; von Arx et al., 2012), although the contribution of these two influences may vary between species. Transects of the flower headspace of 42 species found 30 (71%) produce floral humidity of an intensity greater than would be



nature communications



Article

<https://doi.org/10.1038/s41467-022-35353-8>

A signal-like role for floral humidity in a nocturnal pollination system

Received: 17 April 2022

Accepted: 29 November 2022

Published online: 15 December 2022

Check for updates

Ajinkya Dahake^{1✉}, Piyush Jain², Caleb C. Vogt¹, William Kandalaft¹, Abraham D. Stroock³ & Robert A. Raguso¹

Previous studies have considered floral humidity to be an inadvertent consequence of nectar evaporation, which could be exploited as a cue by nectar-seeking pollinators. By contrast, our interdisciplinary study of a night-blooming flower, *Datura wrightii*, and its hawkmoth pollinator, *Manduca sexta*, reveals that floral relative humidity acts as a mutually beneficial signal in this system. The distinction between cue- and signal-based functions is illustrated by three experimental findings. First, floral humidity gradients in *Datura* are nearly ten-fold greater than those reported for other species, and result from

Despite numerous behavioural studies, a mechanism hasn't yet been proposed.

von Arx et al. (2012)

Harrap et al. (2021)

Dahake et al. (2022)

Overview

Mosquitoes as vectors for disease

Mosquito humidity sensing

Floral humidity and insect pollinators

Mosquitoes as pollinator models

Project plan

Mosquito life cycle

So why mosquitoes?

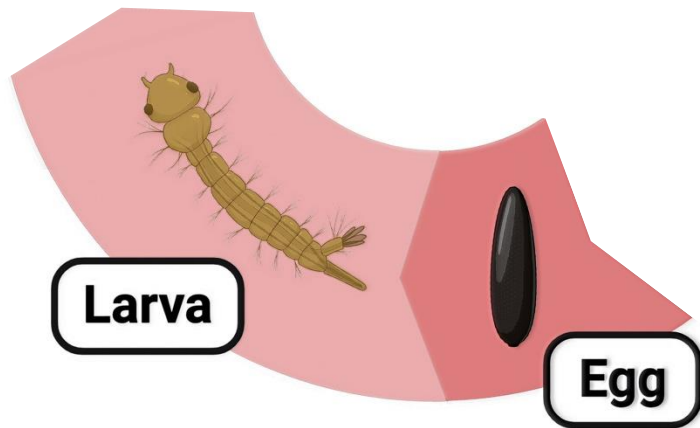
Mosquito life cycle

So why mosquitoes?



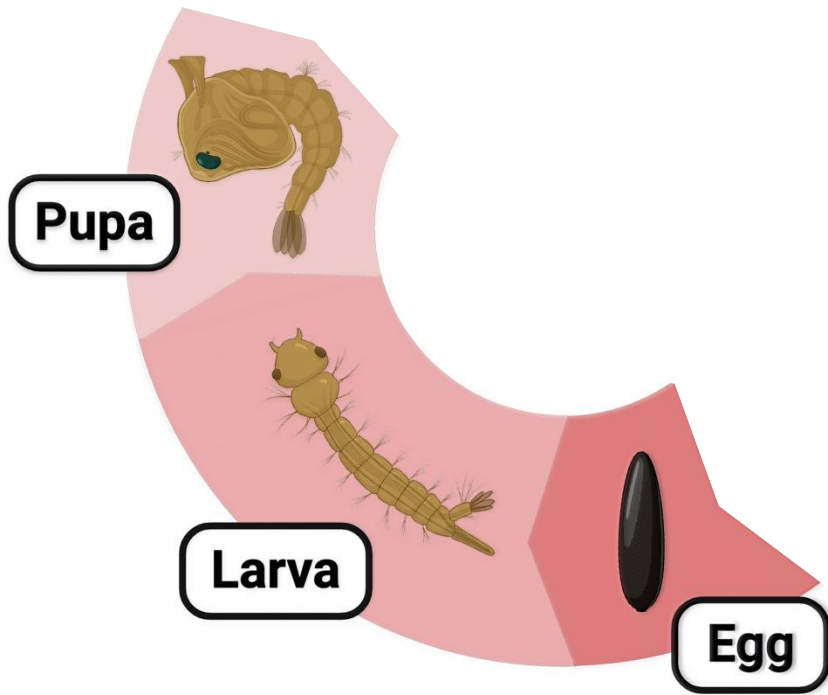
Mosquito life cycle

So why mosquitoes?



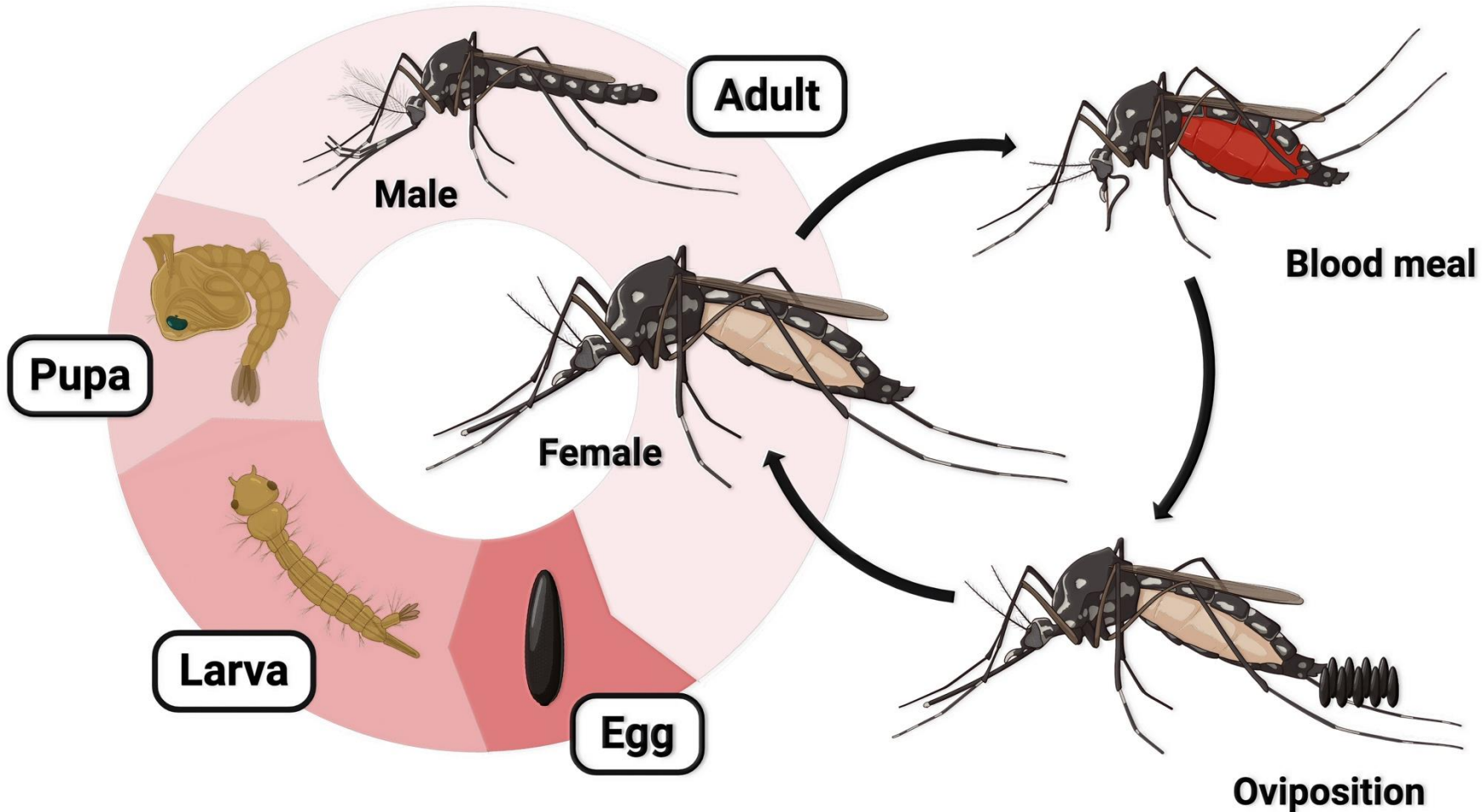
Mosquito life cycle

So why mosquitoes?



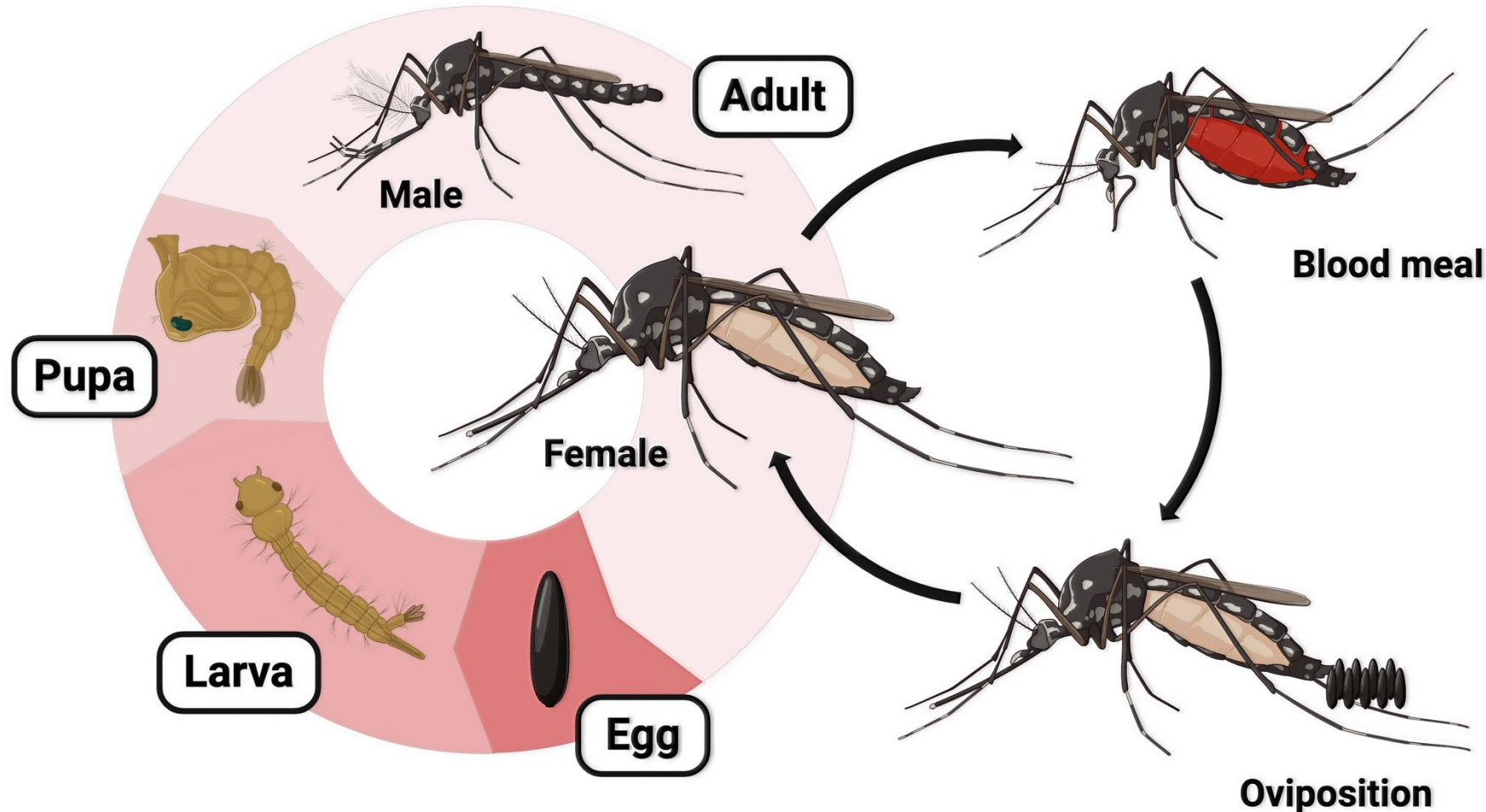
Mosquito life cycle

So why mosquitoes?



Mosquito life cycle

So why mosquitoes?



Hygroreceptor involvement in **gonotrophic cycle** well characterised...

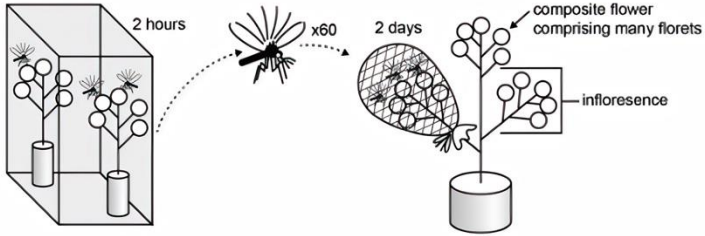
but their role in **nectar-feeding** has is yet to be explored.

Mosquitoes as pollinators

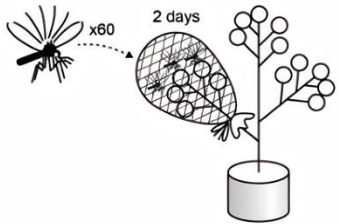
Mosquitoes as pollinators

Pollination experiment #1

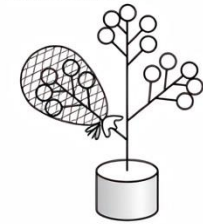
Cross-pollination



Self-pollination

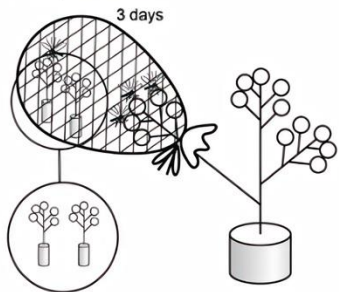


Blank control

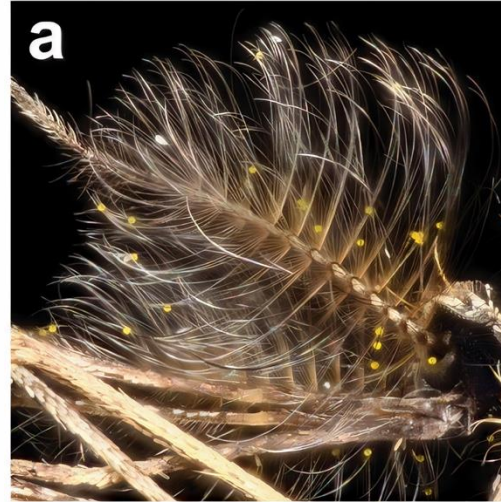
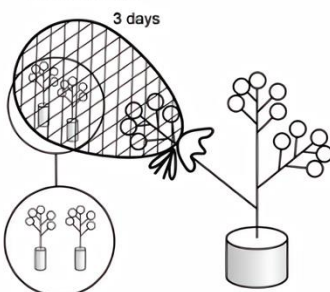


Pollination experiment #2

Cross-pollination



Blank control

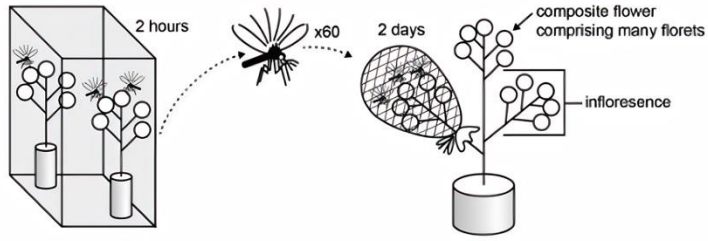


Peach and Gries (2016)

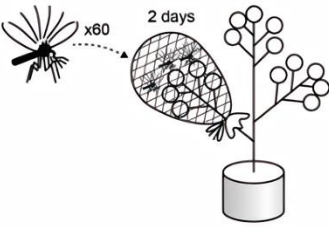
Mosquitoes as pollinators

Pollination experiment #1

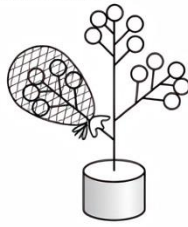
Cross-pollination



Self-pollination

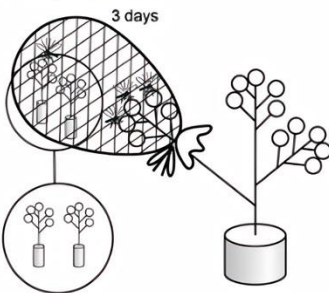


Blank control

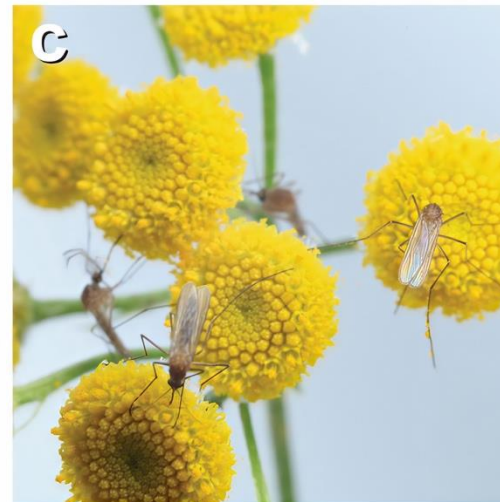
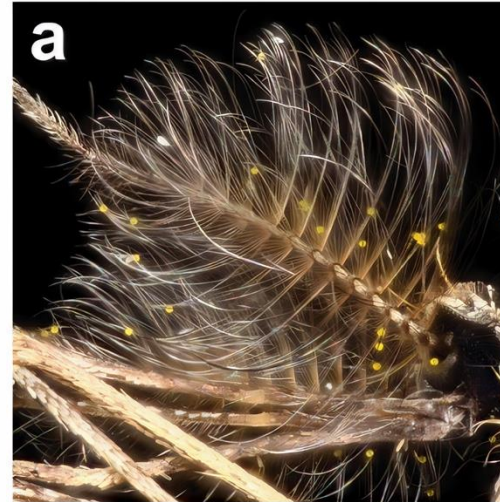
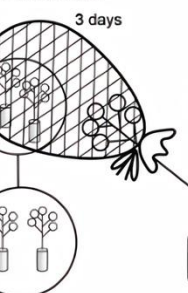


Pollination experiment #2

Cross-pollination



Blank control



The olfactory basis of orchid pollination by mosquitoes

Chloé Lahondère^{a,1}, Clément Vinauer^{a,1}, Ryo P. Okubo^a, Gabriella H. Wolff^a, Jeremy K. Chan^a, Omar S. Akbari^{a,b}, and Jeffrey A. Riffell^{a,2}

^aDepartment of Biology, University of Washington, Seattle, WA 98195; and ^bDivision of Biological Sciences, Section of Cell and Developmental Biology, University of California, San Diego, CA 92093

Edited by Walter S. Leal, University of California, Davis, CA, and accepted by Editorial Board Member John R. Carlson November 22, 2019 (received for review June 20, 2019)

Mosquitoes are important vectors of disease and require sources of carbohydrates for reproduction and survival. Unlike host-related behaviors of mosquitoes, comparatively less is understood about the mechanisms involved in nectar-feeding decisions, or how this sensory information is processed in the mosquito brain. Here we show that *Aedes* spp. mosquitoes, including *Aedes aegypti*, are effective pollinators of the *Platanthera obtusata* orchid, and demonstrate this mutualism is mediated by the orchid's scent and the balance of excitation and inhibition in the mosquito's antennal lobe (AL). The *P. obtusata* orchid emits an attractive, nonanal-rich scent, whereas related *Platanthera* species—not visited by mosquitoes—emit scents dominated by lilac aldehyde. Calcium imaging experiments in the mosquito AL revealed that nonanal and lilac aldehyde each respectively activate the LC2 and AM2 glomeruli, and remarkably, the AM2 glomerulus is also sensitive to N,N-diethyl-methyl-tolamide (DEET), a mosquito repellent. Lateral inhibition between these 2 glomeruli reflects the level of attraction to the orchid scents. Whereas the enriched nonanal scent of *P. obtusata* activates the LC2 and suppresses AM2, the high level of lilac aldehyde in the other orchid scents inverts this pattern of glomerular activity, and behavioral attraction is lost. These results demonstrate the ecological importance of mosquitoes beyond operating as disease vectors and open the door toward understanding the neural basis of mosquito nectar-seeking behaviors.

the flowers, and the importance of mosquito visitation for orchid pollination, are unknown.

In this article, we examine the neural and behavioral processes mediating mosquito floral preference. We present findings from 1) pollination studies in *P. obtusata* by *Aedes* mosquitoes, 2) analyses of floral scent compounds that attract diverse mosquito species, and 3) antennal and antennal lobe (AL) recordings showing how these floral scents and compounds are represented in the mosquito brain (SI Appendix, Fig. S1). Using this integrative approach, we demonstrate that *Aedes* discrimination of *Platanthera* orchids is mediated by the balance of excitation and inhibition in the mosquito antennal lobe.

Results

To understand the importance of various pollinators, including mosquitoes, on *P. obtusata*, we first conducted pollinator observation and exclusion experiments in northern Washington State where *Platanthera* orchids and mosquitoes are abundant. Using a combination of video recordings and focal observations by trained participants, more than 581 *P. obtusata* flowers were observed for a total of 47 h, with 57 floral feeding events by mosquitoes. During our observations, flowers were almost solely visited by various mosquito species (both sexes) that mainly belonged to the *Aedes* group (Fig. 1A and B and SI Appendix, Table S2), with the only

Platanthera | mosquitoes | *Aedes aegypti* | olfaction | nectar

Significance

Mosquitoes are important vectors of disease, such as dengue, malaria, or Zika, and are considered one of the deadliest animal on earth (1); for this reason, research has largely focused on mosquito-host interactions, and in particular, the mosquito's sensory responses to those hosts (2–5). Nectar feeding is one such aspect of mosquito sensory biology that has received comparatively less attention, despite being an excellent system in which to probe the neural bases of behavior (6). For instance, nectar and sugar feeding is critically important for both male and female mosquitoes, serving to increase their lifespan, survival rate, and reproduction, and for males, it is required for survival (6, 7). Mosquitoes are attracted to, and feed on, a variety of plant nectar sources, including those from flowers (8–12). Although most examples of mosquito-plant interactions have shown that mosquitoes contribute little in reproductive services to the plant (13), there are examples of mosquitoes being potential pollinators (9, 10, 14–17). However, few studies have identified the floral cues that serve to attract and mediate these decisions by the mosquitoes and how these behaviors influence pollination.

The association between the *Platanthera obtusata* orchid and *Aedes* mosquitoes is one of the few examples that shows mosquitoes as effective pollinators (14–17) and thus provides investigators a unique opportunity to identify the sensory mechanisms that help mosquitoes locate sources of nectar. The genus *Platanthera* has many different orchid species having diverse morphologies and specialized associations with certain pollinators (SI Appendix, Table S1), with *P. obtusata* being an exemplar with its association with mosquitoes (14–17). Although mosquito visitation has been described in this species (15), the cues that attract mosquitoes to

Nectar feeding by mosquitoes is important for survival and reproduction, and hence disease transmission. However, we know little about the sensory mechanisms that mediate mosquito attraction to sources of nectar, like those of flowers, or how this information is processed in the mosquito brain. Using a unique mutualism between *Aedes* mosquitoes and *Platanthera obtusata* orchids, we reveal that the orchid's scent mediates this mutualism. Furthermore, lateral inhibition in the mosquito's antennal (olfactory) lobe—via the neurotransmitter GABA—is critical for the representation of the scent. These results have implications for understanding the olfactory basis of mosquito nectar-seeking behaviors.

Author contributions: C.L., C.V., R.P.O., G.H.W., J.K.C., and J.A.R. designed research; C.L., C.V., R.P.O., G.H.W., J.K.C., and J.A.R. performed research; O.S.A. contributed new reagents/analytic tools; C.L., C.V., R.P.O., G.H.W., J.K.C., and J.A.R. analyzed data; and C.L., C.V., R.P.O., G.H.W., J.K.C., O.S.A., and J.A.R. wrote the paper.

Competing interest statement: A provisional patent on the odor that mimics the orchid scent was recently filed (SI Appendix, Table S2).

This open access article is distributed under Creative Commons Attribution-NonCommercial-ND 4.0 International license (CC BY-NC-ND).

Data deposition: Data is available on Mendeley Data (<https://data.mendeley.com/datasets/1m42q2p1>).

Present address: Department of Biochemistry, Virginia Polytechnic Institute and State University, Blacksburg, VA 24061.

To whom correspondence should be addressed: Email: jriffell@u.washington.edu.

This article contains supporting information online at <https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.1910589117/-/DCSupplemental>.

First published December 23, 2019.

708-716 | PNAS | January 7, 2020 | vol. 117 | no. 1

www.pnas.org/lookup/doi/10.1073/pnas.1910589117

Lahondère et al. (2020)

Peach and Gries (2016)

Overview

Mosquitoes as vectors for disease

Mosquito humidity sensing

Floral humidity and insect pollinators

Mosquitoes as pollinator models

Project plan

Project plan

Overall aim

Project plan

Overall aim

To determine whether hygrometers are involved in mosquito nectar-foraging.

Project plan

Overall aim

To determine whether hygrometers are involved in mosquito nectar-foraging.

Pilot study

Project plan

Overall aim

To determine whether hygrometors are involved in mosquito nectar-foraging.

Pilot study

1. Obtain *Ae. aegypti* transcriptomic data from NCBI SRA and pre-process using FastQC and fastp.

Project plan

Overall aim

To determine whether hygrometors are involved in mosquito nectar-foraging.

Pilot study

1. Obtain *Ae. aegypti* transcriptomic data from NCBI SRA and pre-process using FastQC and fastp.
2. Assemble *Ae. aegypti* transcriptomes and quantify expression using HISAT2 and featureCounts.

Project plan

Overall aim

To determine whether hygrometors are involved in mosquito nectar-foraging.

Pilot study

1. Obtain *Ae. aegypti* transcriptomic data from NCBI SRA and pre-process using FastQC and fastp.
2. Assemble *Ae. aegypti* transcriptomes and quantify expression using HISAT2 and featureCounts.
3. Identify protein families using OrthoFinder.

Project plan

Overall aim

To determine whether hygrometors are involved in mosquito nectar-foraging.

Pilot study

1. Obtain *Ae. aegypti* transcriptomic data from NCBI SRA and pre-process using FastQC and fastp.
2. Assemble *Ae. aegypti* transcriptomes and quantify expression using HISAT2 and featureCounts.
3. Identify protein families using OrthoFinder.
4. Compare transcriptomic expression of key hygrometors between sexes over their life cycle using DESeq2 in R.

Project plan

Overall aim

To determine whether hygrometors are involved in mosquito nectar-foraging.

Pilot study

1. Obtain *Ae. aegypti* transcriptomic data from NCBI SRA and pre-process using FastQC and fastp.
2. Assemble *Ae. aegypti* transcriptomes and quantify expression using HISAT2 and featureCounts.
3. Identify protein families using OrthoFinder.
4. Compare transcriptomic expression of key hygrometors between sexes over their life cycle using DESeq2 in R.

Expanded study

Project plan

Overall aim

To determine whether hygrometors are involved in mosquito nectar-foraging.

Pilot study

1. Obtain *Ae. aegypti* transcriptomic data from NCBI SRA and pre-process using FastQC and fastp.
2. Assemble *Ae. aegypti* transcriptomes and quantify expression using HISAT2 and featureCounts.
3. Identify protein families using OrthoFinder.
4. Compare transcriptomic expression of key hygrometors between sexes over their life cycle using DESeq2 in R.

Expanded study

1. Expand transcriptomic dataset with additional mosquito species.

Project plan

Overall aim

To determine whether hygrometors are involved in mosquito nectar-foraging.

Pilot study

1. Obtain *Ae. aegypti* transcriptomic data from NCBI SRA and pre-process using FastQC and fastp.
2. Assemble *Ae. aegypti* transcriptomes and quantify expression using HISAT2 and featureCounts.
3. Identify protein families using OrthoFinder.
4. Compare transcriptomic expression of key hygrometors between sexes over their life cycle using DESeq2 in R.

Expanded study

1. Expand transcriptomic dataset with additional mosquito species.
2. Pre-process data, assemble transcriptomes, quantify expression and identify protein families.

Project plan

Overall aim

To determine whether hygrometors are involved in mosquito nectar-foraging.

Pilot study

1. Obtain *Ae. aegypti* transcriptomic data from NCBI SRA and pre-process using FastQC and fastp.
2. Assemble *Ae. aegypti* transcriptomes and quantify expression using HISAT2 and featureCounts.
3. Identify protein families using OrthoFinder.
4. Compare transcriptomic expression of key hygrometors between sexes over their life cycle using DESeq2 in R.

Expanded study

1. Expand transcriptomic dataset with additional mosquito species.
2. Pre-process data, assemble transcriptomes, quantify expression and identify protein families.
3. Normalise expression levels between datasets.

Project plan

Overall aim

To determine whether hygropceptors are involved in mosquito nectar-foraging.

Pilot study

1. Obtain *Ae. aegypti* transcriptomic data from NCBI SRA and pre-process using FastQC and fastp.
2. Assemble *Ae. aegypti* transcriptomes and quantify expression using HISAT2 and featureCounts.
3. Identify protein families using OrthoFinder.
4. Compare transcriptomic expression of key hygropceptors between sexes over their life cycle using DESeq2 in R.

Expanded study

1. Expand transcriptomic dataset with additional mosquito species.
2. Pre-process data, assemble transcriptomes, quantify expression and identify protein families.
3. Normalise expression levels between datasets.
4. Compare transcriptomic expression of hygropceptors across mosquito species as well as that of CO₂ and VOC receptors.

Summary

- Burden of mosquito-borne diseases has resulted in extensive research interest and a vast library of publicly available datasets.

Summary

- Burden of mosquito-borne diseases has resulted in extensive research interest and a vast library of publicly available datasets.
- Mosquitoes locate hosts through a multimodal cascade of sensory cues, including humidity.

Summary

- Burden of mosquito-borne diseases has resulted in extensive research interest and a vast library of publicly available datasets.
- Mosquitoes locate hosts through a multimodal cascade of sensory cues, including humidity.
- Humidity is detected by hygrosensory sensilla, activating moist and dry cells containing ionotropic hygroreceptors.

Summary

- Burden of mosquito-borne diseases has resulted in extensive research interest and a vast library of publicly available datasets.
- Mosquitoes locate hosts through a multimodal cascade of sensory cues, including humidity.
- Humidity is detected by hygrosensory sensilla, activating moist and dry cells containing ionotropic hygroreceptors.
- Elevated humidity is produced by flowering plants around the corolla.

Summary

- Burden of mosquito-borne diseases has resulted in extensive research interest and a vast library of publicly available datasets.
- Mosquitoes locate hosts through a multimodal cascade of sensory cues, including humidity.
- Humidity is detected by hygroscopic sensilla, activating moist and dry cells containing ionotropic hygrosensors.
- Elevated humidity is produced by flowering plants around the corolla.
- Floral humidity acts as a foraging cue for insect pollinators but a mechanism for this hasn't yet been established.

Summary

- Burden of mosquito-borne diseases has resulted in extensive research interest and a vast library of publicly available datasets.
- Mosquitoes locate hosts through a multimodal cascade of sensory cues, including humidity.
- Humidity is detected by hygroscopic sensilla, activating moist and dry cells containing ionotropic hygrosensors.
- Elevated humidity is produced by flowering plants around the corolla.
- Floral humidity acts as a foraging cue for insect pollinators but a mechanism for this hasn't yet been established.
- Mosquitoes offer an exciting opportunity to study this due to their complex life cycle which varies between sexes.

References

- Dahake, A., Jain, P., Vogt, C.C., Kandalaft, W., Stroock, A.D., Raguso, R.A., 2022. A signal-like role for floral humidity in a nocturnal pollination system. *Nat. Commun.* 13, 7773. <https://doi.org/10.1038/s41467-022-35353-8>
- Davis, T.J., Kline, D.L., Kaufman, P.E., 2016. *Aedes albopictus* (Diptera: Culicidae) Oviposition Preference as Influenced by Container Size and *Buddleja davidii* Plants. *J. Med. Entomol.* 53, 273–278. <https://doi.org/10.1093/jme/tjv201>
- Harrap, M.J.M., Hempel De Ibarra, N., Knowles, H.D., Whitney, H.M., Rands, S.A., 2021. Bumblebees can detect floral humidity. *J. Exp. Biol.* 224, jeb240861. <https://doi.org/10.1242/jeb.240861>
- Harrap, M.J.M., Hempel De Ibarra, N., Knowles, H.D., Whitney, H.M., Rands, S.A., 2020. Floral Humidity in Flowering Plants: A Preliminary Survey. *Front. Plant Sci.* 11, 249. <https://doi.org/10.3389/fpls.2020.00249>
- Laursen, W.J., Budelli, G., Tang, R., Chang, E.C., Busby, R., Shankar, S., Gerber, R., Greppi, C., Albuquerque, R., Garrity, P.A., 2023. Humidity sensors that alert mosquitoes to nearby hosts and egg-laying sites. *Neuron* 111, 874-887.e8. <https://doi.org/10.1016/j.neuron.2022.12.025>
- Peach, D.A.H., Gries, G., 2016. Nectar thieves or invited pollinators? A case study of tansy flowers and common house mosquitoes. *Arthropod-Plant Interact.* 10, 497–506. <https://doi.org/10.1007/s11829-016-9445-9>
- Ryan, S.J., Carlson, C.J., Mordecai, E.A., Johnson, L.R., 2019. Global expansion and redistribution of Aedes-borne virus transmission risk with climate change. *PLoS Negl. Trop. Dis.* 13, e0007213. <https://doi.org/10.1371/journal.pntd.0007213>
- Tang, R., Busby, R., Laursen, W.J., T. Keane, G., Garrity, P.A., 2024. Functional dissection of mosquito humidity sensing reveals distinct Dry and Moist Cell contributions to blood feeding and oviposition. *Proc. Natl. Acad. Sci.* 121, e2407394121. <https://doi.org/10.1073/pnas.2407394121>
- van Breugel, F., Riffell, J., Fairhall, A., Dickinson, M.H., 2015. Mosquitoes Use Vision to Associate Odor Plumes with Thermal Targets. *Curr. Biol.* 25, 2123–2129. <https://doi.org/10.1016/j.cub.2015.06.046>
- von Arx, M., Goyret, J., Davidowitz, G., Raguso, R.A., 2012. Floral humidity as a reliable sensory cue for profitability assessment by nectar-foraging hawkmoths. *Proc. Natl. Acad. Sci.* 109, 9471–9476. <https://doi.org/10.1073/pnas.1121624109>

Any questions?